

Notes on Quiver Representations and Quiver Varieties

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Chapter 1

Quiver Representations and Quiver Varieties

1.1 Quiver Representations

1.1.1 Basic definitions and examples

Notation. We use $\mathbf{k}$ to denote the ground field, it could be any field without specification.

Definition 1.1.1. A **quiver** $\vec{Q}$ is a directed graph. It can be described by a set of vertices Q_0 , a set of edges Q_1 , and two maps $s, t : Q_1 \rightarrow Q_0$ which assign to every edge its source and target respectively. We could also use $e : i \rightarrow j$ to indicate that $s(e) = i, t(e) = j$ for the edge e .

Definition 1.1.2. A **representation of quiver** $\vec{Q}$ (over field $\mathbf{k}$) is the collection of data:

- For each vertex $i \in Q_0$, there is a vector space V_i over $\mathbf{k}$ attached to it.
- For each edge $e : i \rightarrow j \in Q_1$, there is a linear map $\mathbf{x}_e : V_i \rightarrow V_j$.

We sometimes simply use the notation $(V, \mathbf{x})$ or V for representation of quiver $\vec{Q}$ without causing ambiguities.

Definition 1.1.3. A **morphism of representations** $f : V \rightarrow W$ between representations $(V, \mathbf{x})$ and $(W, \mathbf{x}')$ of quiver $\vec{Q}$ is a collection of linear maps $(f_i : V_i \rightarrow W_i)_{i \in Q_0}$ such that it commutes with linear operators $\mathbf{x}$, that is to say $f_j \mathbf{x}_e = \mathbf{x}'_e f_i$ for any edge $e : i \rightarrow j \in Q_1$.

$$\begin{array}{ccc} V_i & \xrightarrow{\mathbf{x}_e} & V_j \\ f_i \downarrow & & \downarrow f_j \\ W_i & \xrightarrow{\mathbf{x}'_e} & W_j \end{array}$$

The morphisms of representations $V \rightarrow W$ form a vector space, we use $\text{Hom}_{\vec{Q}}(V, W)$ or simply $\text{Hom}(V, W)$ to denote it if there is no ambiguity. We also use $\text{End}_{\vec{Q}}(V, V)$ to denote the endomorphism algebra of the representation V .

We could say f is invertible if each f_i is, and let $\text{Aut}_{\vec{Q}}(V, V)$ to denote the automorphism group of the representation V .

Definition 1.1.4. We use $\text{Rep}(\vec{Q})$ to denote the category whose objects are those finite dimensional representations of $\vec{Q}$ and morphisms are just morphisms of finite dimensional representations defined above.

The category $\text{Rep}(\vec{Q})$ is endowed with the following operations

- For $(V, \mathbf{x}), (W, \mathbf{x}') \in \text{Rep}(\vec{Q})$, their **direct sum** $(V \oplus W, \mathbf{x} \oplus \mathbf{x}')$ is given by $(V \oplus W)_i := V_i \oplus W_i, (\mathbf{x} \oplus \mathbf{x}')_e := \mathbf{x}_e \oplus \mathbf{x}'_e$.
- We say $V \subseteq W$ is a **subrepresentation** if $V_i \subseteq W_i$ for any $i \in Q_0$ and $\mathbf{x}_e(V_i) \subseteq V_j$ for any $e \in Q_1$, here $\mathbf{x}$ is the linear operator for W . In this situation, the **quotient representation** $(W/V, \bar{\mathbf{x}})$ is given by $(W/V)_i := W_i/V_i$ with the natural quotient map $\bar{\mathbf{x}}$.
- For a morphism $f : V \rightarrow W$ between finite dimensional representations, the **kernel** $\text{Ker } f$ and **image** $\text{Im } f$ of f in $\text{Rep}(\vec{Q})$ are given by $(\text{Ker } f)_i := \text{Ker}(f_i)$ and $(\text{Im } f)_i := \text{Im}(f_i)$ respectively.

Remark. It is easy to check that $\text{Im } f \simeq V/\text{Ker } f$ for $f \in \text{hom}(\text{Rep}(\vec{Q}))$. Thus the category $\text{Rep}(\vec{Q})$ has cokernels, which implies it is an Abelian category (it is clear that $\text{Rep}(\vec{Q})$ is additive).

Definition 1.1.5. Consider a representation $V \in \text{Rep}(\vec{Q})$, it is

- **simple** if it has no nontrivial subrepresentations.
- **semisimple** if it can be written as the direct sum of simple representations.
- **indecomposable** if it cannot be written a direct sum of nontrivial representations.

Example 1.1.6. (i) Let $\vec{Q}$ be the quiver $\bullet \longrightarrow \bullet$ whose underlying graph is the same as dynkin A_2 . It is clear to

see that $\begin{smallmatrix} 0 & \longrightarrow & \mathbf{k} \\ \bullet & & \bullet \\ 1 & & 2 \end{smallmatrix}$ and $\begin{smallmatrix} \mathbf{k} & \longrightarrow & 0 \\ \bullet & & \bullet \\ 1 & & 2 \end{smallmatrix}$ are both simple representations, and in fact the only simple representations. While

$\begin{smallmatrix} \mathbf{k} & \longrightarrow & \mathbf{k} \\ \bullet & & \bullet \\ 1 & & 2 \end{smallmatrix}$ is an indecomposable representation but not a simple one.

(ii) The case when $\vec{Q} = \begin{smallmatrix} \bullet & \rightrightarrows & \bullet \\ 1 & & 2 \end{smallmatrix}$ is more complicated. It is called the Kronecker quiver.

We have $\text{Rep}(\vec{Q})$ to be $\{(V_1, V_2, \mathbf{x}, \mathbf{y}) \mid \text{where } V_1, V_2 \text{ are vector spaces, } \mathbf{x}, \mathbf{y} \text{ linear maps}\}$. When we take both V_1, V_2 to be one-dimensional, the isomorphism classes of representations are parametrized by $\mathbf{k}^2/\mathbf{k}^\times \simeq \{0\} \cup \mathbb{P}^1(\mathbf{k})$.

(iii) Let $\vec{Q} = \begin{smallmatrix} \bullet & \curvearrowright & \bullet \\ & 1 & \end{smallmatrix}$, called the Jordan quiver.

The representations of $\vec{Q}$ is given by the pair $(V, \mathbf{x})$, where $\mathbf{x} : V \rightarrow V$ is a linear map. The isomorphism classes in $\text{Rep}(\vec{Q})$ are given by the dimension of $\mathbf{k}$ -vector space V and conjugacy classes of $\mathbf{x}$. When $\mathbf{k}$ is algebraically closed, it is equivalent to classify by Jordan blocks. This gives the origin of the name of this quiver.

Proposition 1.1.7. Take $\vec{Q}$ to be a quiver without oriented cycles, then any simple representation is of the $\{S(i)\}_{i \in Q_0}$, where

$$S(i)_j = \begin{cases} \mathbf{k}, & i = j \\ 0, & i \neq j \end{cases}$$

Proof. Assume V is a simple representation of $\vec{Q}$, then there is no edge $e \in Q_1$ such that both $V_{s(e)}$ and $V_{t(e)}$ have nonzero dimension since $\vec{Q}$ has no oriented cycle, otherwise, we can have one nontrivial subrepresentation of V by taking the vector space attached on one side to be zero.

Choose one vertex $i \in Q_0$ such that $V_i \neq 0$, we have a subrepresentation V' by letting $V'_i = V_i, V'_j = 0$, for $j \neq i$. Since V is simple, $V = V'$ and hence be of the form $S(i)$. $\square$

We have seen from the Example 1.1.6, (i) that there happens to have some indecomposable representations that could NOT be written as the sum of simple representations (i.e. not semisimple). Thus, it is NOT enough to see only the simple representations. Fortunately, we have the following

Theorem 1.1.8. Any finite-dimensional representation of a quiver $\vec{Q}$ without oriented cycle can be written as a direct sum of indecomposable representations, and is unique up to reordering of sum.

1.1.2 Algebraic Properties of Quiver Representations

To further study the representation category $\text{Rep}(\vec{Q})$ of a quiver $\vec{Q}$, we introduce the path algebra $\mathbf{k}\vec{Q}$ associated with $\vec{Q}$.

Definition 1.1.9. Let $\vec{Q}$ be a quiver. A **path** p of length l in $\vec{Q}$ is a sequence of edges such that the source of each next edge coincides with the target of the previous one:

$$p = (e_l, \dots, e_1), \quad s(e_{i+1}) = t(e_i). \quad \text{Eg: } \dots \xrightarrow{e_{i-1}} \bullet \xrightarrow{e_i} \bullet \xrightarrow{e_{i+1}} \bullet \xrightarrow{e_{i+2}} \dots$$

The natural **concatenation** of paths is given by $(e_1, \dots, e_l) (e'_m, \dots, e'_1) = \begin{cases} (e_1, \dots, e_l, e'_m, \dots, e'_1), & s(e_1) = t(e'_m) \\ 0 & \text{other.} \end{cases}$

We can extend the definition of path of length l to the path of length 0. Define paths $\tilde{p}_i$ to have both source and target to be i with the concatenation law $\tilde{p}_i p = \begin{cases} p, & t(p) = i, \\ 0 & \text{other.} \end{cases}$ and $p \tilde{p}_i = \begin{cases} p, & s(p) = i \\ 0 & \text{other.} \end{cases}$

We define the path algebra $\mathbf{k}\tilde{Q}$ of the quiver $\tilde{Q}$ to be the algebra over $\mathbf{k}$ with basis given by paths in $\tilde{Q}$, added with formal sum, multiplication given by concatenations and with the unit $1 = \sum_{i \in Q_0} \tilde{p}_i$.

Remark. We note that the definition of path algebra is dependent on the orientation of $\tilde{Q}$. By finiteness of the size of quiver $\tilde{Q}$, we know $\mathbf{k}\tilde{Q}$ is a finite dimensional algebra if and only if $\tilde{Q}$ has no oriented cycle, if and only if it is a non-commutative (two-sided) Artin algebra.

Proposition 1.1.10. *There is an equivalence between the category $\mathbf{k}\tilde{Q}\text{-mod}$ of finite dimensional left $\mathbf{k}\tilde{Q}$ -modules and $\text{Rep}(\tilde{Q})$.*

Proof. For $(V, \mathbf{x}) \in \text{Rep}(\tilde{Q})$, we set $V^\vee := \oplus_{i \in Q_0} V_i$. For each path $p \in \mathbf{k}\tilde{Q}$, associate a operator $\mathbf{x}_p$ on $V^\vee$ by setting $\mathbf{x}_{\tilde{p}_i} := \text{Id}_{V_i}$ and $\mathbf{x}_{e_1, \dots, e_l} := \mathbf{x}_{e_l} \circ \dots \circ \mathbf{x}_{e_1}$, where $e_i \in Q_1$. This gives $V^\vee$ a $\mathbf{k}\tilde{Q}$ -module structure.

Conversely, Assume M is a $\mathbf{k}\tilde{Q}$ -module, set $M_i := \tilde{p}_i M, i \in Q_0$, and for each edge $e : i \rightarrow j$, take $\mathbf{x}_e : M_i \rightarrow M_j$ to be same as the operator given by e -action on M . This gives M a $\tilde{Q}$ -representation structure.

By above we construct two functors between $\mathbf{k}\tilde{Q}\text{-mod}$ and $\text{Rep}(\tilde{Q})$, it is not hard to check they are inverse to each other. $\square$

Remark. By Hopkin's theorem, see [ARS95], we know that for a left module of an Artin ring, it is finitely generated if and only if it is both Noetherian and Artinian, and hence it has a finite length composition series. With the equivalence in Proposition 1.1.10, we prove the Theorem 1.1.8. In fact, there are tight connections between representation theory of quivers and so called Auslander-Reiten theory of modules of Artinian algebras, see also [ARS95] for more details.

We next consider projective modules and resolutions. To abbreviate the notation, let $A := \mathbf{k}\tilde{Q}$.

Consider the modules $P(i) := A\tilde{p}_i$. We know projective modules are direct summand of free modules, then $P(i)$ is projective since $A = \oplus_{i \in Q_0} P(i)$. By our settings for path algebra, we know $P(i)$ is the set of linear combinations of paths starting at vertex i .

Proposition 1.1.11. *For any representation V of $\tilde{Q}$, there is an isomorphism $\text{Hom}_{\tilde{Q}}(P(i), V) \simeq V_i$.*

Proof. Notice that regarded as a representation, $P(i)_j = e_j A e_i$ at each vertex j . Fix $v \in V_i$, for each path p starts at i , ends at j , $\mathbf{x}_p(v) \in V_j$. The map between $P(i)$ and V is given by $p \mapsto \mathbf{x}_p(v)$. For edge $e : j \rightarrow k$ and path $p : i \rightarrow j$, since $\mathbf{x}_e(\mathbf{x}_p(v)) = \mathbf{x}_{ep}(v)$, we know it is a homomorphism of representations. The isomorphism as vector spaces then follows. $\square$

Theorem 1.1.12. *Assume that the quiver $\tilde{Q}$ has no oriented cycle, then $\{P(i)\}_{i \in Q_0}$ is the full set of projective elements in $\text{Rep}(\tilde{Q})$.*

Proof. See [Kir16] for a proof. $\square$

Remark. In the language of Grothendieck group, we consider $K(\text{Rep}(\tilde{Q}))$ (we have seen it is an abelian category) as an abelian group with relation $[A] + [C] - [B] = 0$ for short exact sequence in $C_\bullet(\text{Rep}(\tilde{Q}))$.

If $\tilde{Q}$ has no oriented cycle, we have $[P(i)] = \sum_{j \in Q_0} p_{ij} [S(j)]$, where p_{ij} here denotes the number of paths from i to j .

Theorem 1.1.13. *Let $\tilde{Q}$ be a quiver and $A = \mathbf{k}\tilde{Q}$ be its path algebra. Then for any finitely generated $\mathbf{k}\tilde{Q}$ -module V , there is a projective resolution of V called **standard resolution** given by*

$$0 \longrightarrow \bigoplus_{e \in Q_1} P(t(e)) \otimes \mathbf{k}e \otimes V_{s(e)} \xrightarrow{d_1} \bigoplus_{i \in Q_0} P(i) \otimes V_i \xrightarrow{d_0} V \longrightarrow 0$$

where $d_1 : p \otimes e \otimes v \mapsto pe \otimes v - p \otimes \mathbf{x}_e(v)$ and $d_0 : p \otimes v \mapsto \mathbf{x}_p(v)$. The tensor products in above sequence are all over $\mathbf{k}$.

Proof. This comes from the general result in homological algebra. Assume A is an associative algebra with unit 1, M is an A -module. If A_0 is a subalgebra and L is a subspace of A with $A_0 L \subseteq L, L A_0 \subseteq L$ such that A_0, L generates A , then there is an exact sequence of A -modules $A \otimes_{A_0} L \otimes_{A_0} M \xrightarrow{d_1} A \otimes_{A_0} M \xrightarrow{d_0} M \rightarrow 0$, where $d_1 : a \otimes l \otimes m \mapsto al \otimes m - a \otimes lm$ $d_0 : a \otimes m \mapsto am$.

In our case, choose L to be the subspace of A spanned by $e \in Q_1$ and $A_0 := \oplus_{i \in Q_0} \mathbf{k} \tilde{p}_i$. So we can see that $A \otimes_{A_0} V = \oplus_i P(i) \otimes_{(\oplus_i \mathbf{k} \tilde{p}_i)} \oplus_i V_i \simeq \bigoplus_{i \in Q_0} P(i) \otimes V_i$, and similar for $A \otimes_{A_0} L \otimes_{A_0} V \simeq \bigoplus_{e \in Q_1} P(t(i)) \otimes \mathbf{k}(e) \otimes V_{s(e)}$. For the remaining, we need to show the injectivity of d_1 . In fact, d_1 is given by

$$\sum_{e \in Q_1} \sum_{p, s(p)=t(e)} p \otimes e \otimes v_{s(e)} \mapsto \sum_{e, p} (pe \otimes v_{s(e)} - p \otimes \mathbf{x}_e(v_{s(e)}))$$

Assume the image of d_1 is zero, if the left is nonzero, then there is path p of maximal length l with $v_{s(e)} \neq 0$. We thus have $\sum_k p_k e_k \otimes v_k = 0$ since it could NOT be equal to $p' \otimes v$ with p' of strictly lower length, where k is indexed by all paths of maximal length l .

Moreover those $p_k e_k$ are all linearly independent, the only possibility is $v_k = 0$, a contradiction. $\square$

With the Theorem 1.1.13, we know that there are enough projective objects in the category $\text{Rep}(\vec{Q})$, we denote the Ext functor in category $\text{Rep}(\vec{Q})$ by $\text{Ext}_{\vec{Q}}^\bullet$. By comparison theorem of projective resolution, we can easily conclude that

Corollary 1.1.14. *For any $V, W \in \text{Rep}(\vec{Q})$, we have $\text{Ext}_{\vec{Q}}^i(V, W) = 0$ for all $i > 1$. Moreover, the subrepresentation of any projective representation is projective.*

1.1.3 Quadratic Form and Classification of Quivers

For any two representations $V, W \in \text{Rep}(\vec{Q})$, set $\langle \cdot, \cdot \rangle : \text{Rep}(\vec{Q}) \times \text{Rep}(\vec{Q}) \rightarrow \mathbb{Z}$ to be

$$\langle V, W \rangle := \sum_{i \in \mathbb{Z}} (-1)^i \dim \text{Ext}^i(V, W) = \dim \text{Hom}_{\vec{Q}}(V, W) - \dim \text{Ext}_{\vec{Q}}^1(V, W)$$

Theorem 1.1.15. *The value of $\langle V, W \rangle$ only depends on $\mathbf{dim} V := (\dim(V)_i)_{i \in Q_0}$ and $\mathbf{dim} W$.*

Proof. For any short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$, we have $\langle A, W \rangle + \langle C, W \rangle - \langle B, W \rangle = 0$ by definition. Thanks to the standard resolution constructed in Theorem 1.1.13, we can take the resolution of V , abbreviate as $0 \rightarrow P_1 \rightarrow P_0 \rightarrow V \rightarrow 0$ to compute concretely. In fact $\langle V, W \rangle = \langle P_0, W \rangle - \langle P_1, W \rangle = \sum_{i \in Q_0} \dim V_i \dim W_i - \sum_{e \in Q_1} \dim V_{s(e)} \dim W_{t(e)}$ since $\text{Hom}_{\vec{Q}}(P(i), W) = W_i$ from Proposition 1.1.11 and $\text{Ext}_{\vec{Q}}^i(P(i), W) = 0, \forall i > 0$ as $P(i)$ projective. $\square$

Hence, we can extend the form $\langle V, W \rangle$ to $\mathbf{v}, \mathbf{w} \in \mathbb{Z}^{Q_0}$.

Definition 1.1.16. The **Euler form** $\langle \cdot, \cdot \rangle : \mathbb{Z}^{Q_0} \times \mathbb{Z}^{Q_0} \rightarrow \mathbb{Z}$ for a quiver $\vec{Q}$ is defined as

$$\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{i \in Q_0} \mathbf{v}_i \mathbf{w}_i - \sum_{e \in Q_1} \mathbf{v}_{s(e)} \mathbf{w}_{t(e)}, \forall \mathbf{v}, \mathbf{w} \in \mathbb{Z}^{Q_0}.$$

Note that it is a bilinear form but not a symmetric one. We could symmetrize the Euler form to be $(\mathbf{v}, \mathbf{w}) := \langle \mathbf{v}, \mathbf{w} \rangle + \langle \mathbf{w}, \mathbf{v} \rangle$. Call the associated quadratic form $q_{\vec{Q}}(\mathbf{v}) := \frac{1}{2} (\mathbf{v}, \mathbf{v})$ the **Tits form**.

Notice that after symmetrizing, the Tits form will NOT depend on the original orientation of $\vec{Q}$, but only the underlying graph.

Definition 1.1.17. Let Q be a connected graph, say it is of type **Dynkin** if its associated Tits form is positive definite; say it is of type **Euclidean** if its associated Tits form is positive semidefinite.

We could give a whole classification of those special types of quivers.

Theorem 1.1.18. *Let Q be a connected graph, then*

- (i) *It is of type Dynkin if and only if it is the same as one of simply laced Dynkin diagrams A_n, D_n, E_6, E_7, E_8 .*
- (ii) *It is of type Euclidean if and only if it is the same as one of extended simply laced Dynkin diagrams $\hat{A}_n, \hat{D}_n, \hat{E}_6, \hat{E}_7, \hat{E}_8$.*

Proof. See [Kir16] for a detailed explanation. $\square$

Remark. Notice that the classifications are corresponding to ADE classifications of simply laced simple Lie algebras and simply laced affine Kac-Moody Lie algebras respectively.

1.2 Nakajima Quiver Varieties

1.2.1 Mumford's Geometric Invariant Theory

In this section, let $\mathbf{k}$ be a field, we use G to denote a reductive algebraic group over $\mathbf{k}$ and let X to be an affine algebraic variety over $\mathbf{k}$ (i.e. $\text{Spec } \mathbf{k} \rightarrow X$ is separated and of finite type) without specifying.

Definition 1.2.1. Assume that G acts on X , define the **affine GIT quotient** $X // G := \text{Spec } \mathbf{k}[X]^G$, where $\mathbf{k}[X]$ is the ring of regular functions on X and $\mathbf{k}[X]^G$ denotes the G -invariant part.

Remark. Notice there is a natural quotient map $X \rightarrow X // G$ induced by the inclusion $\mathbf{k}[X]^G \hookrightarrow \mathbf{k}[X]$.

Assume that G is linearly reductive, which means every finite-dimensional representation of G is semisimple. By Hilbert's theorem (see [Muk03] Theorem 4.51), we have $\mathbf{k}[X]^G$ to be finitely generated.

We note also that when G is over a field of characteristic zero, then reductive is equivalent to linearly reductive, see for example [Mil17] Theorem 22.138, but it is NOT true in general for non-zero characteristic cases.

We can consider $\mathbb{O} \subseteq X$ a G -orbit in X , it gives a maximal ideal $J_{\mathbb{O}} = \{f \in \mathbf{k}[X]^G \mid f|_{\mathbb{O}} = 0\} \subseteq \mathbf{k}[X]^G$ which corresponding to a closed point in $X // G$. The question then arises is, how to separate G -orbits? Is there G -invariant form to distinct two orbits?

Definition 1.2.2. Two G -orbits $\mathbb{O}, \mathbb{O}' \subset X$ are said to be **closure-equivalent** if there exists a sequence of orbits between them

$$\mathbb{O} = \mathbb{O}_1, \mathbb{O}_2, \dots, \mathbb{O}_{n-1}, \mathbb{O}_n = \mathbb{O}'$$

such that $\overline{\mathbb{O}_i} \cap \overline{\mathbb{O}_{i+1}} \neq \emptyset$ for $i = 1, \dots, n-1$.

Theorem 1.2.3 (Nagata, Mumford). Assume that G is linearly reductive acting on affine variety X , then for two G -orbits $\mathbb{O}$ and $\mathbb{O}'$, the following are equivalent

- (i) The intersection of their closure is non-empty, that is $\overline{\mathbb{O}} \cap \overline{\mathbb{O}'} \neq \emptyset$.
- (ii) $\mathbb{O}$ and $\mathbb{O}'$ are closure-equivalent.
- (iii) $\mathbb{O}$ and $\mathbb{O}'$ could NOT be separated by G -invariant functions in $\mathbf{k}[X]^G$.

Proof. By definition, (i) implies (ii), and since the G -invariant functions are continuous, (ii) implies (iii). It remains to show that (iii) implies (i). Assume that $\overline{\mathbb{O}} \cap \overline{\mathbb{O}'} = \emptyset$, we want to show $\mathbb{O}$ and $\mathbb{O}'$ are separated by $\mathbf{k}[X]^G$.

Let $I := I(\overline{\mathbb{O}})$ and $I' := I(\overline{\mathbb{O}'})$ be the vanishing ideals in $\mathbf{k}[X]$ of the closure $\mathbb{O}$ and $\mathbb{O}'$ respectively. Consider the ideal $I + I' \subseteq \mathbf{k}[X]$ generated by I and I' . By our assumption, $V(I + I') = V(I) \cap V(I')$ is empty. Thus, from Hilbert's Nullstellensatz, we have $I + I' = \mathbf{k}[X]$.

Since G acts algebraically on X , we have the map $\mathbf{k}[X] \rightarrow \mathbf{k}[X] \otimes \mathbf{k}[G]$ to regard $\mathbf{k}[X]$ as G -representation. And as $\overline{\mathbb{O}}, \overline{\mathbb{O}'}$ is preserved under G -action, we have I and I' being G -subrepresentations. There is a homomorphism of G -representations given by

$$I \oplus I' \rightarrow \mathbf{k}[X], \quad (f, f') \mapsto f + f'$$

It is surjective from the previous discussion. By hypothesis, G is linearly reductive. Since finite-dimensional representations are semisimple, for a surjective homomorphism $\phi : V \rightarrow W$ of G -representations, taking the G -invariant spaces on both sides, the map $\phi^G : V^G \rightarrow W^G$ is also surjective. That is to say $(I \cap \mathbf{k}[X]^G) \oplus (I' \cap \mathbf{k}[X]^G) \rightarrow \mathbf{k}[X]^G$ is also surjective. In conclusion, we could find $f \in I \cap \mathbf{k}[X]^G$ and $f' \in I' \cap \mathbf{k}[X]^G$ such that $f + f' = 1$. The function f vanishes on the orbit $\mathbb{O}$ and takes value 1 on $\mathbb{O}'$. $\square$

The proof in the Theorem 1.2.3 can be extracted to give the following property:

Proposition 1.2.4 (Geometric Reductivity Principle). Let G be a linearly reductive group acting on an affine variety X and let W_1, W_2 be disjoint closed G -invariant subsets of X . Then there exists $f \in \mathbf{k}[X]^G$ such that

$$f|_{W_1} = 0, f|_{W_2} = 1.$$

Corollary 1.2.5. *Take the same assumption as the Theorem 1.2.3. Then there is exactly one closed orbit in each closure-equivalence class, moreover, it is contained in the closure of any other orbits in the class.*

Proof. By Theorem 1.2.3, distinct closed G -orbits are separated by $\mathbf{k}[X]^G$, thus each closure-equivalence class contains at most one closed orbit. Let $\mathbb{O}$ be the orbit of minimal dimension in its class. If $\overline{\mathbb{O}} \setminus \mathbb{O}$ is non-empty, then it is closure-equivalent to $\mathbb{O}$ and of lower dimension, which leads to a contradiction on the minimality. Hence the orbit $\mathbb{O}$ we choose is closed.

For any other orbit $\mathbb{O}'$, we know $\mathbb{O} \cap \overline{\mathbb{O}'} = \overline{\mathbb{O}} \cap \overline{\mathbb{O}'} \neq \emptyset$. Then the last point of the corollary is deduced from the fact that $\overline{\mathbb{O}'}$ is the union of orbits. $\square$

Now we can finally describe the topology of the affine GIT quotient space $X // G$.

Corollary 1.2.6. *As a topological space, $X // G = X / \sim$, where the equivalence relation is defined by $x \sim x'$ if and only if $\overline{\mathbb{O}_x} \cap \overline{\mathbb{O}_{x'}} \neq \emptyset$, where $\mathbb{O}_x$ is the G -orbit of x . For $x \in X$, we denote by $[x]$ the corresponding point in $X // G$, and we have an isomorphism*

$$X // G = \{ \text{closed orbits in } X \}, \quad [x] \mapsto \text{unique closed orbit contained in } \overline{\mathbb{O}_x}$$

Example 1.2.7. Take $X = \text{End}(\mathbf{k}^n) \simeq \text{Mat}_{n \times n}(\mathbf{k})$ to be the space of $n \times n$ matrices and $G = \text{GL}_n$ to be the general linear group, then $\mathbf{k}[X] = \mathbf{k}[T_{11}, \dots, T_{nn}]$ and $\mathbf{k}[X]^G \simeq \mathbf{k}[\lambda_1, \dots, \lambda_n]^{S_n}$ since any GL_n -invariant function on X is determined by its values on diagonal matrices. Thus we have $X // G = \text{Spec } \mathbf{k}[\lambda_1, \dots, \lambda_n]^{S_n} \simeq \mathbf{k}^n$.

We shall see that from the previous constructions, the affine GIT quotient is dependent on merely closed orbits, to generalize the conception, we move on to the projective quotient.

Definition 1.2.8. Let G act algebraically on X . To abbreviate, we write $X = \text{Spec } R$. Define a GR -**module** to be a representation of G as well as an R -module M . That is to say, we have a coaction $\mu_M : M \rightarrow M \otimes_{\mathbf{k}} \mathbf{k}[G]$ and this should also be an R -modules homomorphism such that the following diagram commutes:

$$\begin{array}{ccc} R & \xrightarrow{\mu_R} & R \otimes_{\mathbf{k}} \mathbf{k}[G] \\ \downarrow & & \downarrow \\ M & \xrightarrow{\mu_M} & M \otimes_{\mathbf{k}} \mathbf{k}[G] \end{array}$$

In other words, $\mu_M(rm) = \mu_R(r) \cdot \mu_M(m)$ for all $r \in R, m \in M$.

Definition 1.2.9. Use the same assumption as above. Given an R -module M , we say a G -**linearisation** of M is a coaction $\mu_M : M \rightarrow M \otimes_{\mathbf{k}} \mathbf{k}[G]$ that makes M into a GR -module.

Example 1.2.10. Suppose that M is an invertible R -module that admits a G -linearisation, that is, locally free of rank 1 which corresponding to a invertible sheaf, or line bundle. Consider $\text{Sym}(M) = \bigoplus_{k \geq 0} M^{\otimes k}$, it is a commutative algebra over R with G -action on it. And it is finitely generated due to Hilbert's Theorem stated in Remark of 1.2.1 with a natural gradation. We can define $X //_M G := \text{Proj}(\text{Sym}(M)^G)$, called the **projective quotient** from the GR -module M .

Remark. For a geometric and original description of linearisation, one can refer to [MFK94] for G -linearisation of invertible sheaves on schemes and the conditions of G -linearisable. We state here Mumford's criterion for later use.

Let G be a reductive algebraic group acting on a scheme X with $\mathcal{L}$ an invertible sheaf on X . Let $\sigma : G \times X \rightarrow X$ denote the action and $p_2 : G \times X \rightarrow X$ the projection on the second factor, then a G -linearisation of $\mathcal{L}$ consists of an isomorphism $\phi : \sigma^* \mathcal{L} \simeq p_2^* \mathcal{L}$.

Or, in other word, let L be the corresponding line bundle to $\mathcal{L}$, then the G -linearisation is given by $\Sigma = p_2 \circ \Phi$,

$$\begin{array}{ccc} (G \times X) \times_X L & \xrightarrow{p_2^*} & L \\ \downarrow & & \downarrow \pi \\ G \times X & \xrightarrow{p_2} & X \end{array} \quad \begin{array}{ccc} (G \times X) \times_X L & \xrightarrow{\sigma^*} & L \\ \downarrow & & \downarrow \pi \\ G \times X & \xrightarrow{\sigma} & X \end{array}$$

Φ (curved arrow from $(G \times X) \times_X L$ to $(G \times X) \times_X L$)

This lifts G action on X to the G action $\Sigma : G \times L \rightarrow L$ on L .

Example 1.2.11. Assume G acts on X . Let $\chi : G \rightarrow \mathbf{k}^\times$ be a character of G , define

$$\mathbf{k}[X]^{G, \chi^n} = \{f \in \mathbf{k}[X] \mid f(g.x) = \chi^{-n}(g)f(x), \forall x \in X, g \in G\}, \quad \forall n \geq 0$$

Let $A := \bigoplus_{n \geq 0} \mathbf{k}[X]^{G, \chi^n}$, then by definition, the ring A is a graded algebra. If G is linearly reductive (or if we choose the base field $\mathbf{k}$ to be of characteristic zero), The algebra A is finitely generated.

Then we can define the twisted GIT quotient $X //_\chi G$ by character χ to be

$$X //_\chi G = \text{Proj} \left(\bigoplus_{n \geq 0} \mathbf{k}[X]^{G, \chi^n} \right)$$

Since $\mathbf{k}[X]^G = \mathbf{k}[X]^{G, \chi^0} = A_0$ is the zero gradation of A , we then have a projective morphism $X //_\chi G \rightarrow X // G$.

Remark. In fact, the construction in Example 1.2.11 is the special case of Example 1.2.10. If we take the G -linearisation of R to be $R \rightarrow R \otimes_{\mathbf{k}} \mathbf{k}[G], r \mapsto r \otimes \chi^{-1}$, we get precisely $X //_\chi G$ as a projective quotient.

1.2.2 Stability Conditions

According to Remark , fix $\chi : G \rightarrow \mathbf{k}^\times \simeq \mathbb{G}_m$ a character, and consider $L = X \times \mathbf{k}$ to be a trivial line bundle, the lift action of G on X to G on L is given by $g : (x, z) \mapsto (g.x, \chi^{-1}(g)z)$.

Definition 1.2.12. For a point $x \in X$, it is called χ -**semistable** if for any $z \in \mathbf{k}^\times$, the closure of the G orbit of $\hat{x} = (x, z) \in X \times \mathbf{k}$ is disjoint from the zero section, that is to say:

$$\overline{\mathbb{O}}_{\hat{x}} \cap (X \times \{0\}) = \emptyset.$$

We denote the set of χ -semistable points in X by X_χ^{ss} . An G -orbit $\mathbb{O} \subseteq X$ is said to be χ -**semistable** if $\mathbb{O} \subseteq X_\chi^{ss}$.

Proposition 1.2.13. An element $x \in X$ is χ -semistable if and only if there exists $f \in \mathbf{k}[X]^{G, \chi^n}, n \geq 1$, such that $f(x) \neq 0$.

Proof. If $x \in X_\chi^{ss}$, use the reductivity principle Proposition 1.2.4 on $X \times \mathbf{k}$, there is $\hat{f} \in \mathbf{k}[X \times \mathbf{k}]^G$ such that $\hat{f}|_{\overline{\mathbb{O}}_{\hat{x}}} \neq 0, \hat{f}|_{X \times \{0\}} = 0$. Since $\hat{f}$ is G -invariant,

$$g.\hat{f}(x, z) = \hat{f}(g.(x, z)) = \hat{f}(g.x, \chi^{-1}(g)z) = \hat{f}(x, z)$$

and thus $\hat{f}$ should be in the form $\sum_{n \geq 0} f_n(x)z^{-n}$, where $f_n \in \mathbf{k}[X]^{G, \chi^n}$ such that

$$\hat{f}(g.x, \chi^{-1}(g)z) = \sum_n f_n(gx) \chi^{-n}(g^{-1})z^{-n} = \sum_n f_n(g^{-1}g.x)z^{-n}$$

Since $\hat{f}(x, 0) = 0$, we have $f_0 = 0$. Thus there exists $n \geq 1$ such that $f_n(x) \neq 0$.

Conversely, if $f \in \mathbf{k}[X]^{G, \chi^n}, n \geq 1$ such that $f(x) \neq 0$, then we can lift f to $\hat{f} \in \mathbf{k}[X \times \mathbf{k}]$ by setting $\hat{f}(x, z) := f(x)z^{-n}$, clearly $\hat{f}$ is G -invariant and separates $\overline{\mathbb{O}}_{\hat{x}=(x,z)}$ and $X \times \{0\}$ with $z \neq 0$. Then use the same method as in Theorem 1.2.3, we can see x is χ -semistable. $\square$

The corollaries of this Proposition are immediate.

Corollary 1.2.14. (1) $X^{ss} \subset X$ is open and G -invariant (possibly empty).

(2) If $N \in \mathbb{N}^*$, then $x \in X$ is χ^N -semistable if and only if it is χ semistable.

Similar to the consequences of the Theorem 1.2.3, the points $M //_\chi G$ are determined by orbits, which are semistable instead of closed.

Theorem 1.2.15. (i) Let G be a linearly reductive group acting on an affine variety X , and let χ be a character of G . Then two semistable orbits $\mathbb{O}, \mathbb{O}' \subset X^{ss}$ define the same point in $X //_\chi G$ if and only if $\overline{\mathbb{O}} \cap \overline{\mathbb{O}'} \cap X^{ss} \neq \emptyset$ here $\overline{\mathbb{O}}$ is the closure of orbit $\mathbb{O}$ in X .

(ii) As a topological space, $X //_\chi G = X^{ss} / \sim$, where the equivalence relation is given by $x \sim x'$ if and only if $\overline{\mathbb{O}}_x \cap \overline{\mathbb{O}}_{x'} \cap X^{ss} \neq \emptyset$. For $x \in X^{ss}$ we denote by $[x]$ the corresponding point in $X //_\chi G$.

Definition 1.2.16. Take the same assumption as Theorem 1.2.15, an element $x \in X^{ss}$ is called **stable** if the stabilizer $G_x \subseteq G$ is finite and the orbit $\mathbb{O}_x$ is closed in X^{ss} .

Remark. (1) The definition of stable point here is the same as the definition of **pre-stable** in Mumford's original definition, see [MFK94] Definition 1.7. Without the finiteness condition on stabilizer, the point would be stable in Mumford's notation.

(2) When $x \in X^{ss}$ is semisimple with G_x finite, we have the following to be equivalent:

- (i) The orbit $\mathbb{O}_x$ is closed in X^{ss} .
- (ii) For any $z \in \mathbf{k}^\times$, the orbit of $\hat{x} = (x, z) \in X \times \mathbf{k}$ is closed in $M \times \mathbf{k}$.
- (iii) There exists $f \in \mathbf{k}[X]^{G, \chi^n}$, $n \geq 1$, such that $f(x) \neq 0$ and the action of G in $A_f = \{x \in X^{ss} \mid f(x) \neq 0\}$ is closed.

1.2.3 GIT Quotient of Quiver representations

In this section, we may choose the base field $\mathbf{k}$ to be algebraically closed for convenience.

Recall that for a quiver $\vec{Q}$, the representation of $\vec{Q}$ is $(V, \mathbf{x})$ is given by the vector spaces attached onto each vertex and linear operators $\mathbf{x}_e$ for edges $e \in Q_1$.

Now, we choose $\mathbf{v} = \dim V$ the dimension vector, we can identify $V_i \simeq \mathbf{k}^{\mathbf{v}_i}$ for $i \in Q_0$. Then the representation of $\vec{Q}$ is determined by the collection $\mathbf{x}_e \in \text{Hom}_{\mathbf{k}}(\mathbf{k}^{\mathbf{v}_i}, \mathbf{k}^{\mathbf{v}_j})$, $e : i \rightarrow j \in Q_1$, or equivalently, determined by a vector $\mathbf{x}$ in the space

$$\mathcal{R}(\vec{Q}, \mathbf{v}) = \bigoplus_{e \in Q_1} \text{Hom}_{\mathbf{k}}(\mathbf{k}^{\mathbf{v}_{s(e)}}, \mathbf{k}^{\mathbf{v}_{t(e)}})$$

Conversely, every $\mathbf{x} \in \mathcal{R}(\vec{Q}, \mathbf{v})$ defines a representation $V^{\mathbf{x}} = (\{\mathbf{k}^{\mathbf{v}_i}\}, \{\mathbf{x}_e\})$. We call $\mathcal{R}(\vec{Q}, \mathbf{v})$ the **representation space**.

Consider the group $\text{GL}(\mathbf{v}) = \prod_{i \in Q_0} \text{GL}(\mathbf{v}_i)$ acts on the representation space $\mathcal{R}(\vec{Q}, \mathbf{v})$ by conjugation, for $g = (g_i)_{i \in Q_0} \in \text{GL}(\mathbf{v})$,

$$\begin{array}{ccc} V_i & \xrightarrow{g \cdot \mathbf{x}_e} & V_j \\ g_i \downarrow & & \uparrow g_j^{-1} \\ V_i & \xrightarrow{\mathbf{x}_e} & V_j \end{array}$$

Since the subgroup $\mathbf{k}^\times \subseteq \text{GL}(\mathbf{v})$ (note that $\mathbf{k}^\times$ embedded in $\text{GL}(\mathbf{v})$ diagonally) acts trivially, The action is actually given by $\text{PGL}(\mathbf{v}) = \text{GL}(\mathbf{v})/\mathbf{k}^\times$. It is transitive since $\text{GL}(\mathbf{v}_i)$ is transitive over V_i .

Assume $\mathbf{x} \in \mathcal{R}(\mathbf{v})$, the orbit of $V^{\mathbf{x}}$ in the representation space is

$$\mathbb{O}_{V^{\mathbf{x}}} = \text{PGL}(\mathbf{v})/G_{\mathbf{x}}, \quad G_{\mathbf{x}} = \{g \in \text{GL}(\mathbf{v}) \mid g(\mathbf{x}) = \mathbf{x}\} \text{ is the stabilizer of } \mathbf{x}$$

We can consider now the GIT quotient $\mathcal{R}_0(\mathbf{v}) = \mathcal{R}(\mathbf{v}) // \text{PGL}(\mathbf{v})$.

Proposition 1.2.17. Let $0 \rightarrow U \rightarrow V \rightarrow W \rightarrow 0$ be a short exact sequence of $\vec{Q}$ representations, then $\mathbb{O}_{U \oplus W} \subseteq \overline{\mathbb{O}}_V$.

Proof. We use $\mathbf{u}, \mathbf{v}, \mathbf{w}$ to denote the dimension vector. By the exact sequence given, $\mathbf{v} = \mathbf{u} + \mathbf{w} = \dim(U \oplus W)$. Moreover,

$V_i \simeq U_i \oplus W_i$, $i \in Q_0$ as vector space. Thus for each $e : i \rightarrow j \in Q_1$, the linear operator $\mathbf{x}_e^{\mathbf{v}}$ is of the form $\begin{pmatrix} \mathbf{x}_e^{\mathbf{u}} & \mathbf{x}_e' \\ 0 & \mathbf{x}_e^{\mathbf{w}} \end{pmatrix}$.

Set one parameter subgroup in $\text{GL}(\mathbf{v})$ to be $\mathbf{k}^\times \rightarrow \text{GL}(\mathbf{v})$, $t \mapsto \begin{pmatrix} t \cdot 1_{\mathbf{u}} & 0 \\ 0 & 1_{\mathbf{w}} \end{pmatrix} =: g(t)$. Then,

$$\lim_{t \rightarrow 0} g(t) \mathbf{x}_e^{\mathbf{v}} g(t)^{-1} = \lim_{t \rightarrow 0} \begin{pmatrix} \mathbf{x}_e^{\mathbf{u}} & t \mathbf{x}_e' \\ 0 & \mathbf{x}_e^{\mathbf{w}} \end{pmatrix} = \begin{pmatrix} \mathbf{x}_e^{\mathbf{u}} & 0 \\ 0 & \mathbf{x}_e^{\mathbf{w}} \end{pmatrix}$$

Hence the closure of $\overline{\mathbb{O}}_{V^{\mathbf{x}}}$ contains the representation $\begin{pmatrix} \mathbf{x}_e^{\mathbf{u}} & 0 \\ 0 & \mathbf{x}_e^{\mathbf{w}} \end{pmatrix}$ corresponding to $U \oplus W$. □

Lemma 1.2.18. Let $F(V) = V = V^{\geq 0} \supset V^{\geq 1} \supset \dots \supset V^{\geq l} = \{0\}$ be any filtration of V and $\text{gr}^F V = \bigoplus V^{\geq n}/V^{\geq n+1}$ be the graded ring with respect to the filtration. Assume V and V' are two representations of $\vec{Q}$ with the same dimension vector, we have

- (1) If $V' \simeq \text{gr}^F V$ for some filtration F , then $\mathbb{O}_{V'} \subset \overline{\mathbb{O}}_V$.
(2) If $\mathbb{O}_{V'}$ is closed, then the converse also holds, i.e. if $\mathbb{O}_{V'} \subset \overline{\mathbb{O}}_V$, then $V' \simeq \text{gr} V$ for some filtration F .

Note that the filtration of V can be chosen as the composition series $V = W^0 \supset W^1 \supset \dots \supset W^n = \{0\}$, where W^n/W^{n+1} is simple. This is called the **semisimplification** of V . Denote the resulted graded ring by $V^{ss} = \bigoplus W^n/W^{n+1}$.

Theorem 1.2.19. *With the same assumption as above, we have an orbit $\mathbb{O}_V$ being closed if and only if V is semisimple.*

Proof. From the Lemma 1.2.18, we know that the orbit $\mathbb{O}_V$ is closed if and only if for any filtration F of V , $V \simeq \text{gr}^F V$. Apply this to the semisimplification, we know $V \simeq V^{ss}$, hence V can be written as the sum of simple representations. Conversely, if V is semisimple, any filtration of V is given by some composition series which are pairwise equivalent. $\square$

Corollary 1.2.20. *There is a bijection between $\mathcal{R}_0(\mathbf{v})$ and the set of isomorphism classes of semisimple representations of $\vec{Q}$.*

Proof. It is a direct consequence by combining Theorem 1.2.19 and Corollary 1.2.6. $\square$

Further, we could consider the projective GIT quotient by determining the twisting character.

Definition 1.2.21. Let $\theta \in \mathbb{Z}^{Q_0}$, we could define a character χ_θ associated to θ by $\chi_\theta : \text{GL}(\mathbf{v}) \rightarrow \mathbf{k}^\times, g = (g_i)_{i \in Q_0} \mapsto \prod_{i \in Q_0} \det(g_i)^{-\theta_i}$.

Call parameter θ the **stability parameter**.

To ensure the compatibility with $\text{PGL}(\mathbf{v})$, we need $\prod_i \det(\lambda g_i)^{\theta_i} = \prod_i \lambda^{-\mathbf{v}_i \theta_i} \det(g_i) = \prod_i \det(g_i)$, which means $\mathbf{v} \cdot \theta = \sum_i \mathbf{v}_i \theta_i = 0$.

We now could construct the projective GIT quotient to be $\mathcal{R}_\theta(\mathbf{v}) = \mathcal{R}(\mathbf{v}) //_{\chi_\theta} \text{PGL}(\mathbf{v})$. The stability condition will be discussed later.

Recall that for a finite vector space V , its cotangent bundle T^*V is $V \oplus V^*$, with a symplectic form ω on T^*V given as

$$\omega((v_1, \lambda_1); (v_2, \lambda_2)) = \langle \lambda_1, v_2 \rangle - \langle \lambda_2, v_1 \rangle, \quad \forall v_i \in V, \lambda_i \in V^*$$

For the representation space $\mathcal{R}(\vec{Q}, \mathbf{v})$, its cotangent bundle will be denoted by $\mathcal{R}(Q^\sharp, \mathbf{v})$, given by

$$\mathcal{R}(Q^\sharp, \mathbf{v}) = \mathcal{R}(\vec{Q}, \mathbf{v}) \oplus \mathcal{R}(\vec{Q}, \mathbf{v})^* = \bigoplus_{e \in Q_1} \text{Hom}_{\mathbf{k}}(\mathbf{k}^{\mathbf{v}_{s(e)}}, \mathbf{k}^{\mathbf{v}_{t(e)}}) \oplus \text{Hom}_{\mathbf{k}}(\mathbf{k}^{\mathbf{v}_{t(e)}}, \mathbf{k}^{\mathbf{v}_{s(e)}})$$

The dual space of $\text{Hom}(V, W)$ is $\text{Hom}(W, V)$ with the pairing $\langle f, g \rangle = \text{Tr}_V(gf) = \sum_{i=1}^{\dim V} \langle gf(e_i), e_i \rangle_V$. Then there is a symplectic form ω_{Q_1} on $\mathcal{R}(Q^\sharp, \mathbf{v})$ given by

$$\omega_{Q_1}(\mathbf{p}_1, \mathbf{p}_2) = \omega_{Q_1}(\mathbf{x}_1 + \mathbf{x}_1^*, \mathbf{x}_2 + \mathbf{x}_2^*) = \langle \mathbf{x}_1^*, \mathbf{x}_2 \rangle - \langle \mathbf{x}_2^*, \mathbf{x}_1 \rangle, \quad \mathbf{x}_i \in \mathcal{R}(\vec{Q}, \mathbf{v}), \mathbf{x}_i^* \in \mathcal{R}(\vec{Q}^{opp}, \mathbf{v}) = \mathcal{R}(\vec{Q}, \mathbf{v})^*, \mathbf{p}_i = (\mathbf{x}_i, \mathbf{x}_i^*) \in \mathcal{R}(Q^\sharp, \mathbf{v})$$

Note that it is in fact the representation space of double quiver, but the symplectic form is related to the orientation of $\vec{Q}$.

In general, let $Q_1^\sharp$ denote the edge set of doubled quiver $Q^\sharp$, and let $\varepsilon : Q_1^\sharp \rightarrow \mathbf{k}$ be a skew-symmetric function on $Q_1^\sharp$, i.e. $\varepsilon(\bar{e}) = -\varepsilon(e)$ where $\bar{e}$ is the reverse edge of e . We can define the symplectic form associated to ε by

$$\omega_\varepsilon(\mathbf{p}, \mathbf{q}) = \text{Tr}_V \sum_{e \in Q_1^\sharp} \varepsilon(e) \mathbf{p}_{\bar{e}} \mathbf{q}_e, \quad \mathbf{p}, \mathbf{q} \in \mathcal{R}(Q^\sharp, \mathbf{v})$$

If ε is compatible with the orientation of $\vec{Q}$, $\varepsilon(e) = 1, \forall e \in Q_1$, then $\omega_\varepsilon = \omega_{Q_1}$.

Let (M, ω) be a symplectic manifold, with G reductive Lie group acting on M and preserves the symplectic form. Let $\mathfrak{g}$ be the Lie algebra of G , the action of G gives an infinitesimal action of $\mathfrak{g}$, that is, a Lie algebra homomorphism from $\mathfrak{g}$ to symplectic vector fields on M , note by $a \mapsto \xi_a$, satisfying $L_{\xi_a}(\omega) = 0$.

Definition 1.2.22. With the same assumption as above, for any $a \in \mathfrak{g}$, the function H_a such that $\xi_a = X_{H_a}$ is called the **Hamiltonian**.

And the G -symplectic action on M is called **Hamiltonian** if there exists a G -equivariant map $\mu : M \rightarrow \mathfrak{g}^*$ such that

(i) The function $H_a : x \rightarrow \langle \mu(x), a \rangle$ is the Hamiltonian function to $a \in \mathfrak{g}$.

(ii) For any $a, b \in \mathfrak{g}$, we have $\{H_a, H_b\} = H_{[a, b]}$.

The map μ is called the **moment map**. See for example [CG10], Chapter 1.4 for a detailed description.

Theorem 1.2.23. *Consider the action of $\mathrm{GL}(\mathbf{v})$ on $\mathcal{R}(Q^\#, \mathbf{v})$. It is a Hamiltonian action and preserves the symplectic form ω_ε . The corresponding moment map $\mu_{\mathbf{v}}$ is given by the composition*

$$\begin{aligned} \mathcal{R}(Q^\#, \mathbf{v}) &\rightarrow \mathfrak{gl}(\mathbf{v}) = \bigoplus_{i \in Q_0} \mathfrak{gl}(\mathbf{v}_i, \mathbf{k}), \quad \mathbf{p} \mapsto \bigoplus_i \sum_{t(e)=i} \varepsilon(e) z_e z_{\bar{e}} \\ \mathfrak{gl}(\mathbf{v}) &\xrightarrow{\sim} \mathfrak{gl}(\mathbf{v})^*, \quad x \mapsto (\langle x, \cdot \rangle : y \mapsto \mathrm{Tr}(xy)) \end{aligned}$$

Proof. For a symplectic space (W, ω) with G -symplectic action, the Hamiltonian of $a \in \mathfrak{g}$ is given by

$$H_a(x) = \frac{1}{2} \omega(x, a.x) = \langle \mu(x), a \rangle$$

Use this consequence on the doubled representation space, we have

$$\langle \mu_{\mathbf{v}}(\mathbf{p}), a \rangle = \frac{1}{2} \omega_\varepsilon(\mathbf{p}, a.\mathbf{p}) = \frac{1}{2} \sum_{e \in Q^\#} \varepsilon(e) \mathrm{Tr}(\mathbf{p}_{\bar{e}} [a, \mathbf{x}_e]) = \frac{1}{2} \sum_{e \in Q^\#} \varepsilon(e) \mathrm{Tr}(a [\mathbf{p}_e, \mathbf{p}_{\bar{e}}]) = \sum_{e \in Q^\#} \varepsilon(e) \mathrm{Tr}(a \mathbf{p}_e \mathbf{p}_{\bar{e}}).$$

Hence $\mu_{\mathbf{v}}(\mathbf{p}) = \sum_{e \in Q^\#} \varepsilon(e) \mathbf{p}_e \mathbf{p}_{\bar{e}} = \bigoplus_i \sum_{t(e)=i} \varepsilon(e) \mathbf{p}_e \mathbf{p}_{\bar{e}}$. □

Remark. Note that if we choose the symplectic form $\omega = \omega_{Q_1}$ to be along the orientation, write $\mathbf{p} = (\mathbf{x}, \mathbf{x}^*)$, then

$$\mu_{\mathbf{v}}(\mathbf{p}) = \sum_{e \in Q_1} [\mathbf{x}_e, \mathbf{x}_e^*]$$

Definition 1.2.24. Assume $\vec{Q}$ a quiver with $Q^\#$ its doubled quiver, $\mathbf{v} \in \mathbb{Z}_+^{Q_0}$ the dimension vector. Let $\mu_{\mathbf{v}} : \mathcal{R}(Q^\#, \mathbf{v}) \rightarrow \mathfrak{gl}(\mathbf{v})^*$ be the moment map as in Theorem 1.2.23.

Define the **moduli GIT quotient** to be $\mathcal{M}_0(\mathbf{v}) := \mu_{\mathbf{v}}^{-1}(0) // \mathrm{PGL}(\mathbf{v})$.

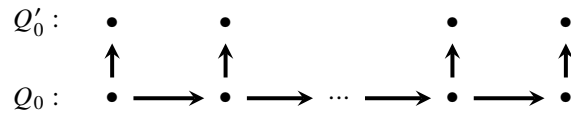
And the **projective GIT moduli** is $\mathcal{M}_\theta(\mathbf{v}) := \mu_{\mathbf{v}}^{-1}(0) //_{\chi_\theta} \mathrm{GL}(\mathbf{v})$. If $\mathbf{v} \cdot \theta = 0$, then $\mathcal{M}_\theta(\mathbf{v}) = \mu_{\mathbf{v}}^{-1}(0) //_{\chi_\theta} \mathrm{PGL}(\mathbf{v})$.

1.2.4 Construction of Nakajima Quiver Varieties

Before giving the definition of quiver varieties, we need to introduce another fundamental operation called framing.

Definition 1.2.25. Fix a quiver $\vec{Q} = (Q_0, Q_1)$, we define the **framing** of $\vec{Q}$ to be the quiver $\vec{Q}^\heartsuit$ with vertex set $Q_0^\heartsuit = Q_0 \amalg Q'_0$ where Q'_0 is another copy of Q_0 . and the edge set $Q_1^\heartsuit = Q_1 \amalg Q'_1$, where Q'_1 are those edges from $i \in Q_0$ to the same point $i \in Q'_0$.

The image of the framed quiver for example is as the following:



Definition 1.2.26. Let $W = \prod_{i \in Q'_0} W_i$ be a collection of vector spaces attached to framing points.

A **representation of the framed quiver** $\vec{Q}^\heartsuit$ or **W -framed representation** of $\vec{Q}$ is a representation $(V_i, \mathbf{x}_e, \mathbf{j}_i)$ where $(V_i, \mathbf{x}_e)$ is the data of representation of $\vec{Q}$, together with a collection of linear maps $\mathbf{j}_i : V_i \rightarrow W_i$.

The **morphism** between two W -framed representations $f : (V, \mathbf{x}, \mathbf{j}) \rightarrow (V', \mathbf{x}', \mathbf{j}')$ is a morphism of representations of $\vec{Q}$ that also satisfies $\mathbf{j}_i = \mathbf{j}'_i \circ f_i$.

Let $\mathbf{v}, \mathbf{w}$ be the dimension vectors of representation V and the framing representation W . Define the representation space of framed representation to be:

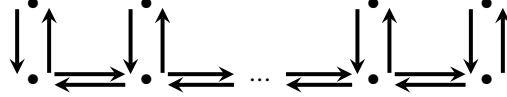
$$\mathcal{R}(\vec{Q}^\heartsuit, \mathbf{v}, \mathbf{w}) = \left(\bigoplus_{i \in I} \text{Hom}_{\mathbf{k}}(\mathbf{k}^{v_i}, \mathbf{k}^{w_i}) \right) \oplus \left(\bigoplus_{e \in Q_1} \text{Hom}_{\mathbf{k}}(\mathbf{k}^{v_{s(e)}}, \mathbf{k}^{v_{t(e)}}) \right)$$

The group $\text{GL}(\mathbf{v})$ acting on $\mathcal{R}(\mathbf{v}, \mathbf{w})$ also by conjugation, and the affine GIT quotient is $\mathcal{R}_0(\mathbf{v}, \mathbf{w}) = \mathcal{R}(\mathbf{v}, \mathbf{w}) // \text{GL}(\mathbf{v})$.

Since $\mathbf{k}^\times$ acting non-trivially on $\mathcal{R}(\mathbf{v}, \mathbf{w})$, we draw back the condition $\theta \cdot \mathbf{v} = 0$, and define $\mathcal{R}_\theta(\mathbf{v}, \mathbf{w}) = \mathcal{R}(\mathbf{v}, \mathbf{w}) //_{\chi_\theta} \text{GL}(\mathbf{v})$.

Now we could combining the operation of framing and Hamiltonian reduction together to get Nakajima quiver variety.

Consider the framed representation of a double quiver, whose image is as:



The representation space is defined as

$$\begin{aligned} \mathcal{R}((Q^\heartsuit)^\sharp, \mathbf{v}, \mathbf{w}) &= \left(\bigoplus_{i \in I} \text{Hom}_{\mathbf{k}}(\mathbf{k}^{v_i}, \mathbf{k}^{w_i}) \right) \oplus \left(\bigoplus_{i \in I} \text{Hom}_{\mathbf{k}}(\mathbf{k}^{w_i}, \mathbf{k}^{v_i}) \right) \oplus \left(\bigoplus_{e \in Q_1^\sharp} \text{Hom}_{\mathbf{k}}(\mathbf{k}^{v_{s(e)}}, \mathbf{k}^{v_{t(e)}}) \right) \\ &= \mathcal{R}(Q^\heartsuit, \mathbf{v}, \mathbf{w}) \oplus \mathcal{R}(Q^\heartsuit, \mathbf{v}, \mathbf{w})^* = T^*(\mathcal{R}(Q^\heartsuit, \mathbf{v}, \mathbf{w})) \end{aligned}$$

With the symplectic form $\omega_\varepsilon(\mathbf{p}, \mathbf{i}, \mathbf{j}; \mathbf{p}', \mathbf{i}', \mathbf{j}') = \text{Tr}_V \left(\sum_{e \in Q_1^\sharp} \varepsilon(e) \mathbf{p}_e \mathbf{p}'_e + \sum_{k \in Q_0} (\mathbf{i}_k \mathbf{j}'_k - \mathbf{i}'_k \mathbf{j}_k) \right)$.

With the symplectic structure, we have the consequence similar to the Theorem 1.2.23, we state it without a detailed proof.

Theorem 1.2.27. *Consider the action of $\text{GL}(\mathbf{v})$ on $\mathcal{R}((Q^\heartsuit)^\sharp, \mathbf{v}, \mathbf{w})$. It is a Hamiltonian action and preserves the symplectic form ω_ε . The corresponding moment map $\mu_{\mathbf{v}, \mathbf{w}}$ is given by the composition of Lie algebra dual map with the following*

$$\mathcal{R}((Q^\heartsuit)^\sharp, \mathbf{v}) \rightarrow \mathfrak{gl}(\mathbf{v}), \quad (\mathbf{p}, \mathbf{i}, \mathbf{j}) \mapsto \sum_{e \in Q_1^\sharp} \varepsilon(e) \mathbf{p}_e \mathbf{p}_e - \sum_{k \in Q_0} \mathbf{i}_k \mathbf{j}_k$$

Definition 1.2.28. Use the same notation as above, we define the **Nakajima quiver variety** by

$$\mathcal{M}_\theta(\mathbf{v}, \mathbf{w}) := \mu_{\mathbf{v}, \mathbf{w}}^{-1}(0) //_{\chi_\theta} \text{GL}(\mathbf{v})$$

and more generally

$$\mathcal{M}_{\lambda, \theta}(\mathbf{v}, \mathbf{w}) := \mu_{\mathbf{v}, \mathbf{w}}^{-1}(\lambda) //_{\chi_\theta} \text{GL}(\mathbf{v})$$

Chapter 2

The Tutte Polynomial and Cell Decomposition of Toric Quiver Varieties

2.1 The Tutte Polynomial and Toric Quiver Varieties

2.1.1 Some Graph Theory

In the next sections, we would like to study on a special case of quiver varieties, called toric varieties, and relate their geometry with graph theory by following [AMV19]. We start with introducing some conceptions in graph theory.

Definition 2.1.1. By **tree** of a graph Γ , we mean a connected acyclic subgraph. By **spanning tree**, we mean a tree that reaches all vertices of the graph.

Definition 2.1.2. Define the **Tutte Polynomial** for a graph Γ with edge set E and vertex set V to be:

$$\mathbf{T}_{\Gamma}(x, y) = \sum_{D \subseteq E} (x - 1)^{k(D) - k(E)} (y - 1)^{k(D) + \#D - \#V}.$$

It is shown by Tutte that the polynomial has no negative coefficient, and can be expressed in the sum over the spanning tree T of the graph Γ .

$$\mathbf{T}_{\Gamma}(x, y) = \sum_T x^{\text{int}(T)} y^{\text{ext}(T)}$$

where $\text{int}(T, \Gamma)$ and $\text{ext}(T, \Gamma)$ are integral weights attached to the spanning tree $T \subseteq \Gamma$, called **internal activity** and **external activity** respectively.

One specialisation of the Tutte polynomial is as the following:

$$\mathbf{T}_{\Gamma}(1, q) = \sum_{D \subseteq E \text{ connected}} (q - 1)^{1 + \#D - \#V} = \sum_T q^{\text{ext}(\Gamma, T)}.$$

The external activity could be computed by using deletion/contraction operation recursively.

For the deletion operation, we consider some edge $e \in E$, deleting it just means removing it from the edge set and keeping the vertices unchanged. For the contraction operation, we consider some non-loop edge $e \in E$, contracting it means sticking two vertices e connects to one point, and shrinking the edge e .

We choose an ordering on the edge set E , and start with the biggest non-loop edge $e \in E$, then the external activity is given by the following algorithm

$$\text{ext}(\Gamma, T) = \begin{cases} \text{ext}(\Gamma \setminus e, T) & \text{if } e \notin T \\ \text{ext}(\Gamma / e, T / e) & \text{if } e \in T \end{cases}$$

Here $\Gamma \setminus e$ is for deleting e and Γ / e for contracting e . The recursion ends when the graph has only one vertex and maybe several loops on it, we then set $\text{ext}(\Gamma, T) = \#\tilde{E}$, where $\tilde{E}$ is the edge set of the resulting graph.

2.1.2 Contraction/Deletion for Quivers and Stability conditions

Notation. We use the same notation as in the previous sections. Fix a quiver $\vec{Q}$, let $\mathbf{v}$ denote the dimension vector of representation and $\mathcal{R}(Q^\sharp, \mathbf{v})$ the corresponding representation space of double quiver $Q^\sharp$ of $\vec{Q}$. The reductive group $G_{\mathbf{v}} = \mathrm{GL}(\mathbf{v})$ acts on $\mathcal{R}(Q^\sharp, \mathbf{v})$. Choose $\mu_{\mathbf{v}} : \mathcal{R}(Q^\sharp, \mathbf{v}) \rightarrow \mathfrak{gl}(\mathbf{v})^*$ the moment map with respect to the orientation on $\vec{Q}$. Then we can consider the GIT moduli quotient

$$\mathcal{M}_{\lambda, \theta}(\mathbf{v}) := \mu_{\mathbf{v}}^{-1}(\lambda) //_{\chi_\theta} G_{\mathbf{v}}$$

We hereby clarify some terminology, recall that the Nakajima quiver variety need dimension vector $\mathbf{w}$ for framing, however, in Crowley-Boevey's sense, the quiver variety is isomorphic to the one obtained from adding one single vertex ∞ of dimension 1 and connecting $i \in Q_0$ to ∞ by $\mathbf{w}_i$, see [Gin09]. In such sense, the GIT moduli quotient in above notation is regarded as a quiver variety that dealing the case of dimension vector $\mathbf{v}$ with any framing $\mathbf{w}$.

For parameters $(\lambda, \theta) \in \mathbf{k}^{Q_0} \times \mathbb{R}^{Q_0}$ determining the GIT quotient, they are said to be **hyperkähler parameters** for $\mathbf{v}$ if they satisfy $\mathbf{v} \cdot \theta = 0, \mathbf{v} \cdot \lambda = 0$.

For hyperkähler parameters λ, θ for $\mathbf{v}$, they are said to be **generic** if any dimension vector $\mathbf{v}' \geq 0$ such that $\mathbf{v} - \mathbf{v}' \geq 0, \mathbf{v}' \cdot \lambda = 0$ and $\mathbf{v}' \cdot \theta = 0$ implies $\mathbf{v}' = \mathbf{v}$ or $\mathbf{v}' = 0$. A nice property that generic parameters provide is that the quiver variety will be smooth, one can refer to [Kuz01] Theorem 2.1 and Lemma 2.11 for some geometrical background.

We first illustrate here the connection between the properties of representation and stability conditions of points, as a completion to the previous content. One major useful result is Mumford's criterion Theorem 2.1.4.

Definition 2.1.3. Let $\theta \in \mathbb{R}^{Q_0}$. Then $V \in \mathrm{Rep}(\vec{Q})$ is θ -semistable (θ -stable resp.) if $\theta \cdot \dim V = 0$ and $\forall V' \subseteq V$ subrepresentation, we have $\theta \cdot \dim V' \leq 0$ ($V' \subseteq V$ properly, $\theta \cdot \dim V' < 0$ resp).

Theorem 2.1.4 (Mumford's numerical criterion). *Let G be a reductive group acting on an affine variety M , and χ be a character of G . $\lambda : \mathbf{k}^\times \rightarrow G$ one-parameter subgroup of G , define $\langle \lambda, \chi \rangle$ to satisfy $t^{\langle \lambda, \chi \rangle} = \chi(\lambda(t))$. Then $x \in M$ is χ -semistable (resp. stable) if and only if for any one-parameter subgroup λ such that the limit $\lim_{t \rightarrow 0} \lambda(t) \cdot x$ exists, we have $\langle \lambda, \chi \rangle \geq 0$ (resp. $\langle \lambda, \chi \rangle > 0$)*

Theorem 2.1.5. *Let $\mathbf{v} \in \mathbb{Z}_+^{Q_0}$, $\theta \in \mathbb{Z}^{Q_0}$, be such that $\theta \cdot \mathbf{v} = 0$. Then $\mathbf{x} \in \mathcal{R}(\mathbf{v})$ is χ_θ -semistable (resp. stable) if and only if $V^{\mathbf{x}}$ is θ -semistable (resp. θ -stable).*

Proof. Abbreviate $V^{\mathbf{x}}$ as V . Assume that V is θ -semistable and $\lambda : \mathbb{C}^\times \rightarrow \mathrm{PGL}(\mathbf{v})$ a one-parameter subgroup such that $\lim_{t \rightarrow 0} \lambda(t) \cdot x$ exists. This gives a weight decomposition $V = \bigoplus_{n \in \mathbb{Z}} V^n$, and the filtration will be

$$\{0\} = V^{\geq b} \subseteq \dots \subseteq V^{\geq a+1} \subseteq V^{\geq a} = V$$

where each $V^{\geq n} = \bigoplus_{m \geq n} V^m$ is a $\mathbf{x}$ -stable subspace, i.e. a subrepresentation. We compute

$$\chi_\theta(\lambda(t)) = \prod_{i \in Q_0} \det(\lambda(t))_i^{-\theta_i} = \prod_{i, n \in \mathbb{Z}} (t^{n \dim(V^n)_i})^{-\theta_i} = t^{-\sum n \theta \cdot \dim V^n}$$

It gives $\langle \chi, \lambda \rangle = -\theta \cdot \mathbf{v}'$, where $\mathbf{v}' := \sum_n n \dim V^n$.

For now $\sum_{m=a}^b \dim V^{\geq m} = \sum_{n=a}^b (n - a + 1) \dim V^n = (1 - a) \dim V + \sum_n n \dim V^n = (1 - a) \mathbf{v} + \mathbf{v}'$.

Since $\theta \cdot \mathbf{v} = 0$, we know

$$\langle \chi, \lambda \rangle = -\theta \cdot \mathbf{v}' = -\sum_{m=a}^b \theta \cdot \dim V^{\geq m}.$$

Each $V^{\geq m}$ is a subrepresentation, while V is θ -semistable. Hence $\theta \cdot \dim V^{\geq m} \leq 0$ for any $m \in [a, b]$ which means $\langle \chi, \lambda \rangle \geq 0$. $\mathbf{x}$ is χ_θ -semistable. (For the stable case, the difference will be $\theta \cdot \dim V^{\geq m} < 0$ for $\forall m \in (a, b]$.)

Conversely, if $\mathbf{x}$ is χ_θ -semistable, then for $V' \subseteq V$ subrepresentation, we could construct one-parameter subgroup $\lambda : \mathbb{C}^\times \rightarrow \mathrm{PGL}(\mathbf{v})$ by setting $\lambda(t) = t$ on V' and $\lambda(t) = 1$ on V/V' . Then by 2.1.4, we know $\langle \chi, \lambda \rangle \geq 0$. Through our computation above, we know this implies $\dim V' \cdot \theta \leq 0$. For the stable case, just choose proper subspace and $\langle \chi, \lambda \rangle > 0$. $\square$

Remark. The above Theorem connects geometric stability with the stability in the representation theory meaning. Notice that the χ_θ -stability is defined for $\theta \in \mathbb{Z}^{Q_0}$ while the stability for representations is defined for $\mathbb{R}^{Q_0}$.

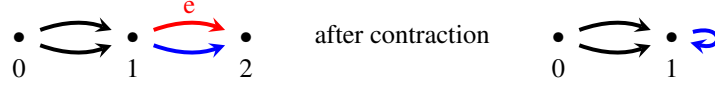
Moreover, we have the similar result for framed and doubled framed quivers.

By a **toric quiver variety** we mean a quiver variety with an action of torus. Equivalently, it is the GIT moduli quotient with dimension vector $\mathbf{v} = (1, \dots, 1)$. In such a case, $G_{\mathbf{v}} = \prod \mathbf{k}^{\times} \simeq \prod \mathbb{G}_m$ is an algebraic torus.

We now restrict ourself to this toric case. Taking $\mathbf{v} = (1, \dots, 1)$ once for all and we omit the parameter $\mathbf{v}$ for abbreviation.

We could also do deletion/contraction on a quiver $\vec{Q}$. To write concretely, for a non-loop edge $e \in Q_1$, $(\vec{Q}/e)_0 = (Q_0 \setminus \{s(e), t(e)\}) \cup \{\iota\}$ and $(\vec{Q}/e)_1 = Q_1 \setminus e$. Consider α the inclusion $(\vec{Q}/e)_1$ in Q_1 and $\beta : Q_0 \rightarrow (\vec{Q}/e)_0$ taking both $s(e), t(e)$ to ι . The source and target maps $s', t' : (\vec{Q}/e)_1 \rightarrow (\vec{Q}/e)_0$ are given by $s' := \beta \circ s \circ \alpha, t' := \beta \circ t \circ \alpha$.

There is an example picture of contraction of a quiver



The parameters λ, θ to determine the quiver varieties are changed under contraction. For $\lambda \in \mathbf{k}^{Q_0}$, we define

$$\lambda/e \in \mathbf{k}^{(\vec{Q}/e)_0} = \begin{cases} \lambda(i) & \text{if } i \in (Q/e)_0 \setminus \{\iota\} \\ \lambda(s(e)) + \lambda(t(e)) & \text{if } i = \iota \end{cases}$$

Given $\theta \in \mathbb{Z}^{Q_0}$, we define $\theta/e \in \mathbb{Z}^{(\vec{Q}/e)_0}$ in a similar manner.

Now we turning into the side of representation theory.

Definition 2.1.6. A representation over field $\mathbf{k}$ is said to be completely indecomposable if it is indecomposable over the algebraic closure $\bar{\mathbf{k}}$.

Let $m(\vec{Q}, q, \mathbf{v})$ denote the number of isomorphism classes of completely indecomposable representations of the quiver $\vec{Q}$ over $\mathbb{F}_q$ with dimension vector $\mathbf{v}$.

Notation. In [Kac83], Kac showed that the number m does NOT depend on the orientation of Q , and can be expressed in polynomial form, we call the **Kac polynomial**, denoted by $A_{\mathbf{v}}(q)$.

Now an immediate application of graph theory on the quiver representation is that the Kac polynomial for toric case coincides with the specialisation of the Tutte polynomial.

Definition 2.1.7. For an element $\mathbf{x} \in \mathcal{R}(\vec{Q})$, set $D := \{e \in Q_1 | \mathbf{x}_e \text{ is not an isomorphism}\}$. Define the **inversion graph** $K_{\mathbf{x}}$ of $\mathbf{x}$ to be the subgraph $Q \setminus D$, where Q is the underlying graph of $\vec{Q}$.

Lemma 2.1.8. The corresponding representation $V^{\mathbf{x}}$ of $\mathbf{x}$ is indecomposable if and only if the inversion graph $K_{\mathbf{x}}$ is connected.

Proof. Since in the toric case, the vector space at each vertex is $\mathbf{k}$, and for each edge $e : i \rightarrow j$, if $\mathbf{x}_e$ is NOT an isomorphism, it will be a zero map.

If e is the only edge connecting these two vertices, then locally it decomposes as the direct sum of $\begin{matrix} 0 & \longrightarrow & \mathbf{k} \\ \bullet & & \bullet \\ i & & j \end{matrix}$ and $\begin{matrix} \mathbf{k} & \longrightarrow & 0 \\ \bullet & & \bullet \\ i & & j \end{matrix}$.

Removing the edge e will make $K_{\mathbf{x}}$ unconnected. If there are other edges connecting vertices i and j such that their corresponding linear operator is non-zero, then locally it could NOT be decomposed, and removing e will NOT affect the connectedness of $K_{\mathbf{x}}$. $\square$

Theorem 2.1.9. $A_{\mathbf{v}}(q) = \sum_{T \subseteq Q} q^{\text{ext } T}$, where $\mathbf{v} = (1, \dots, 1)$.

Proof. By the above Lemma 2.1.8, we can assume that the underlying graph Q of the quiver is connected.

With an ordering on the edge set, let e be the biggest non-loop edge.

If $e \in K_{\mathbf{x}}$, then $\mathbf{x}_e$ is a linear operator that identify $V_{s(e)}$ and $V_{t(e)}$. The contraction will eliminate $s(e)$ and $t(e)$, then produce new vertex ι . With the isomorphism $V_{s(e)} \simeq V_{t(e)}$, we can recover $\mathbf{x}$ from $\mathbf{x}/e$ by split V_{ι} into two equal pieces with an isomorphic linear operator connecting them. Then the correspondence $\mathbf{x} \leftrightarrow \mathbf{x}/e$ is one-to-one up to isomorphism.

If $e \notin K_{\mathbf{x}}$, then it is zero map and we could consider $\mathbf{x}$ as the representation of $Q \setminus e$. Repeat the process until only one point left. Let T be the tree formed by contracted edges, then by definition of external activity, the number of loops at the end of recursion is $\text{ext}(T)$. Representations associated to such tree T are in bijection with representations of the quiver with one vertex and

$\text{ext}(T)$ loops. Take the base field of these representations to be $\mathbb{F}_q$, the number of indecomposable representations hence become $q^{\text{ext}(T)}$.

If T and T' are two different spanning trees, they differ at least one non-loop edge, which gives two different classes of indecomposable representations. Finally, since $\mathbf{v}$ is toric dimension vector, then indecomposable is the same as completely indecomposable. Summing over the spanning trees in the graph Q , we get the number of representation classes we want. $\square$

Corollary 2.1.10. *The Kac polynomial $A_{\mathbf{v}}(q)$ is equal to the specialisation $\mathbf{T}_Q(1, q)$ of Tutte polynomial.*

Definition 2.1.11. For $\mathbf{p} = (\mathbf{x}, \mathbf{x}^*) \in \mathcal{R}(Q^\#)$ a point in the representation space of double quiver, set

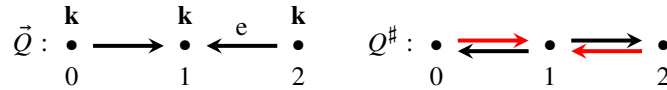
$$D_{\mathbf{p}} := \{e \in Q_1 \mid \text{neither } \mathbf{x}_e \text{ nor } \mathbf{x}_e^* \text{ is an isomorphism}\}$$

and $K_{\mathbf{p}} \subseteq \vec{Q}$ the **inversion graph for $\mathbf{p}$** defined as $\vec{Q} \setminus D_{\mathbf{p}}$.

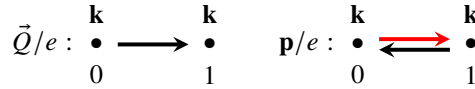
Now for $e \in K_{\mathbf{p}}$ a non-loop edge, we define the contraction $\mathbf{p}/e \in \mathcal{R}(Q^\# / e)$ corresponding to the representation $\mathbf{p}$. Without loss of generality, assume that $\mathbf{x}_e$ is an isomorphism, then set V_i remains unchanged for $i \in (\vec{Q}/e)_0 \setminus \{\iota\}$ and V_ι at the new vertex ι to be

$$V_\iota := \{(v, \mathbf{x}_e(v)) \in V_{s(e)} \oplus V_{t(e)} \mid v \in V_{s(e)}\} \simeq V_{s(e)} \simeq V_{t(e)}.$$

For those edges who meets $s(e)$ or $t(e)$, say $e' \in (\vec{Q}/e)_1$, $\mathbf{x}_{e'}$ and $\mathbf{x}_{e'}^*$ are defined in a natural way with the above isomorphism. Also note that the dimension vector $\mathbf{v}$ will still be toric after the contraction, that is $\mathbf{v} = (1, \dots, 1)$ with length decreasing by 1. The following graphs give a simplest example for the contraction.



Above is the original quiver and representations, the red edges indicate the orientation such that $\mathbf{x}_h$ are isomorphisms. After contraction, they become:



Lemma 2.1.12. Note by $\mathbf{Z}_\lambda = \mu_{\mathbf{v}}^{-1}(\lambda)$ the locus of μ^{-1} at λ . Let $e \in K_{\mathbf{p}}$ be a non-loop edge, if $\mathbf{p} \in \mathbf{Z}_\lambda$, then $\mathbf{p}/e \in (\mathbf{Z}/e)_{(\lambda/e)}$.

Proposition 2.1.13. Assume $\mathbf{p}$ is θ -stable and $e \in K_{\mathbf{p}}$ is a non-loop edge, then $\mathbf{p}/e$ is θ/e -stable.

Proof. Assume that $\mathbf{p}/e$ is (θ/e) -unstable, let $\mathbf{q}/e \subseteq \mathbf{p}/e$ be a destabilising submodule. We can lift $\mathbf{q}/e$ to $\mathcal{R}(Q^\#)$.

- If the vector space of $\mathbf{q}/e$ at ι is full-dimensional, then the lifted vector spaces at $s(e), t(e)$ are chosen to be $V_{s(e)} = \mathbf{k}$ and $V_{t(e)} = \mathbf{k}$.
- If the vector space of $\mathbf{q}/e$ at ι is 0, then lift to $V_{s(e)} = V_{t(e)} = 0$.

Notice such a lift gives a submodule $\mathbf{q}$ of $\mathbf{p}$ and can be contracted exactly to $\mathbf{q}/e$. We have

$$\theta \cdot \dim V^{\mathbf{q}} = \sum_{i \in (Q/e)_0 \setminus \{\iota\}} \theta_i \dim V_i^{\mathbf{q}/e} + (\theta_{s(e)} + \theta_{t(e)}) \dim V_{s(e)} \geq 0$$

Use the result of Theorem 2.1.5, this implies that $\mathbf{p}$ is NOT θ -stable, a contradiction to the assumption. $\square$

Proposition 2.1.14. For $\theta \in \mathbb{Z}^{Q_0}$, if $\mathbf{p} = (\mathbf{x}, \mathbf{x}^*) \in \mathcal{R}(Q^\#)$ is θ -stable, then $K_{\mathbf{p}}$ is a connected graph.

Proof. Suppose if $K_{\mathbf{p}}$ is not connected. Choose a connected component of $K_{\mathbf{p}}$, say K' , with vertices set $K'_0 \subseteq Q_0$. Let $\delta : Q_0 \rightarrow \mathbb{R}$ be the indication map for K'_0 . There is a non-trivial (since K' is non-empty) one-parameter subgroup $(t^{\delta(i)} \cdot I_{\mathbf{v}_i})_{i \in Q_0} \subseteq \mathbf{G}_{\mathbf{v}}$ whose action stabilises $\mathbf{x}$ (we choose the orientation for $\mathbf{x}$ such that $(\mathbf{x}_h)_{h \in K'_1}$ are isomorphisms). Moreover, such stabilizer subgroup is proper and of positive dimension in the torus $\mathbf{G}_{\mathbf{v}}$. This contradicts to the stability condition for $\mathbf{p}$ (recall we need two condition for stability, one is the finiteness of stabilizer group and the other is the 'closedness' of the orbit). $\square$

By above proposition, we know that if $\mathbf{p} \in \mathcal{R}(Q^\#)$ is θ -stable, then $K_{\mathbf{p}}$ contains a spanning tree. In fact, if $K_{\mathbf{p}}$ contains no spanning tree, there is a vertex in Q_0 not reached by edges of $K_{\mathbf{p}}$, but $K_{\mathbf{p}} = Q \setminus D_{\mathbf{p}}$ only loses edges with all vertices preserved. Hence such isolated vertex will be a connected component in $K_{\mathbf{p}}$, a contradiction to the consequence of Proposition 2.1.14 that $K_{\mathbf{p}}$ is connected.

2.1.3 Cell Decomposition of Toric Quiver varieties

We shall see that the graphic components of the underlying graph of quiver $\vec{Q}$ (i.e. trees) could be translated into topological components of toric quiver varieties (i.e. cell decompositions).

Consider now a spanning tree T in the graph Q of the quiver, we could attach an orientation on T with respect to the generic parameter θ .

For an edge $e \in T$ (neglect its orientation in the quiver $\vec{Q}$), we could remove it to get two connected components contained in T , say T_1^e and T_2^e . If $\sum_{i \in T_1^e} \theta_i < \sum_{j \in T_2^e} \theta_j$, take e to be oriented from T_1^e to T_2^e and vice versa.

Remark. Notice $\sum_{i \in T_1^e} \theta_i$ shall not be equal to $\sum_{j \in T_2^e} \theta_j$. Otherwise $\sum_{i \in T_1^e} \theta_i = \sum_{j \in T_2^e} \theta_j = 0$, which contradicts to stability condition.

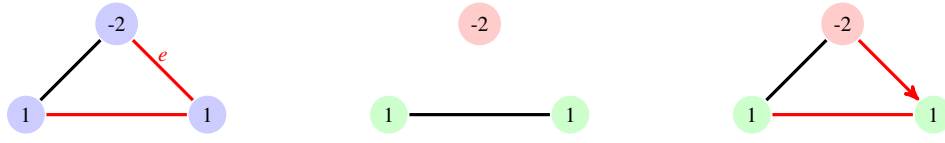


Figure 2.1: Determine the orientation of one tree in $\tilde{A}_2$

The above figure shows how to give orientation for selected edge (e) in a spanning tree T (marked in red) with respect to a fixed stability parameter θ (marked as number at each node).

We denote by $T^\theta \subset Q^\#$ the oriented spanning tree defined by θ induced orientation. Denote by $T_{s(e)}^\theta$ and $T_{t(e)}^\theta$ the two connected component determined by edge e with its new orientation.

Definition 2.1.15. For $\mathbf{p} \in \mathcal{R}(Q^\#)$, set $D^\# := \{e \in Q_1^\# \mid \mathbf{x}_e \text{ is not an isomorphism}\}$. Define the **oriented inversion graph** $K_{\mathbf{p}}^\theta$ of $\mathbf{p}$ to be $Q^\# \setminus D^\#$. Notice that $K_{\mathbf{p}}^\theta \subseteq (K_{\mathbf{p}})^\# \subseteq Q^\#$.

Proposition 2.1.16. Let $\theta \in \mathbb{R}^{Q_0}$ and $\mathbf{p} \in \mathcal{R}(Q^\#)$. Then $\mathbf{p}$ is θ -stable if and only if there $\exists T \subseteq Q$ spanning tree such that $T^\theta \subseteq K_{\mathbf{p}}^\theta$.

Proof. Assume first that $\mathbf{p}$ is θ -stable, suppose by contradiction such T^θ does NOT exist. With Proposition 2.1.14, we know $K_{\mathbf{p}}$ contains at least one spanning tree. Take any spanning tree $T \subseteq K_{\mathbf{p}}^\theta$, there is an edge $e \in T^\theta$ that $e \notin K_{\mathbf{p}}^\theta$. Let $Q_{s(e)}$ and $Q_{t(e)}$ denote the subgraph of Q corresponding to $T_{s(e)}^\theta, T_{t(e)}^\theta$ respectively. Since $e \notin K_{\mathbf{p}}^\theta$, both arrows of e are zero map, then we could construct a subrepresentation of $\mathbf{p}$ such that the vector spaces U_j are the same as those V_j of $\mathbf{p}$ for $j \in (Q_{s(e)})_0$ and 0 for those $j \in (Q_{t(e)})_0$. By the relation $\sum_{i \in (Q_{s(e)})_0} \theta_i > \sum_{j \in (Q_{t(e)})_0} \theta_j$ and $\sum_{i \in (Q_{s(e)})_0} \theta_i + \sum_{j \in (Q_{t(e)})_0} \theta_j = 0$, we know $\sum_{i \in (Q_{s(e)})_0} \theta_i > 0$. It gives a destabilising subrepresentation of $\mathbf{p}$, a contradiction.

Conversely, if there exists $T^\theta \subseteq K_{\mathbf{p}}^\theta$ and assume $\mathbf{p}$ is unstable. Let $\mathbf{q} \subseteq \mathbf{p}$ be a destabilising subrepresentation. If $\mathbf{q}$ is a decomposable representation, then one of its decomposition summand is destabilising subrepresentation of $\mathbf{p}$. We thus reduce to the case when $\mathbf{q}$ is indecomposable. Consider the subgraph K of T such that $\mathbf{q}$ is non-zero at its vertices. If K is not connected, then there must be vertices in T of number greater than zero between any two neighboured connected components of K . This gives a decomposition of $\mathbf{q}$ which contradicts to our previous assumption. Hence K is connected and $T \setminus K$ decomposes into several connected components, denoted by $K_1, \dots, K_m$.

The representation $\mathbf{p}/\mathbf{q}$ is unstable, and the decomposition $T \setminus K$ divides it into direct sum of m representations, one of these subrepresentations is destabilising, denoted by $\mathbf{r}$. Suppose that $\mathbf{r}$ is supported on component K_i , consider $\mathbf{q}' := \text{Ker}(\mathbf{p} \rightarrow \mathbf{r})$. $\mathbf{q}'$ is a subrepresentation of $\mathbf{p}$ supported on $K' := K \cup (\bigcup_{j \neq i, j=1, \dots, m} K_j)$. In T^θ , there is exactly one edge start from each K_i to K and no edge between K_i and K_j , thus K_i connect to K' . However, by destabilising condition $\sum_{w \in K_i} \theta_w > 0 > \sum_{v \in K'} \theta_v$, the orientation hence goes from K' to K_i , a contradiction. $\square$

Recall the discussion of stability condition in doing contraction/deletion, we know if $e \in K_{\mathbf{p}}$, the contraction preserves stability, and if $e \notin K_{\mathbf{p}}$, the deletion preserves stability. If one of $\mathbf{x}_e, \mathbf{x}_e^*$ is isomorphism and the other to be zero, we could choose to do either contraction or deletion. But the deletion would destroy the stability for sometimes, we shall next deal with the case. Consider $\mathbf{p} \in \mathcal{R}(Q^\#)$ being θ -stable, e a non-loop edge. Assume that $\mathbf{p} \setminus e$ is θ -unstable.

Lemma 2.1.17. *Let $\mathbf{p}, e$ be as above, then any destabilising subrepresentation of $\mathbf{p} \setminus e$ must be non-zero at one of $s(e), t(e)$, but NOT both.*

And if one of the destabilising subrepresentations contains $s(e)$, then all of the rest must.

Proof. For the first point, if $\mathbf{q} \setminus e$ a subrepresentation of $\mathbf{p} \setminus e$ that vanishes at neither $s(e)$ nor $t(e)$, lift $\mathbf{q} \setminus e$ to $\mathbf{q}$ with the triviality at e will give a destabilising subrepresentation of $\mathbf{p}$. Otherwise if $\mathbf{q} \setminus e$ having the vector spaces at both two vertices $s(e)$ and $t(e)$ to be non-zero, we could also lift to a subrepresentation $\mathbf{q} \subseteq \mathbf{p}$ by identifying the edge e , this is again a destabilising subrepresentation of $\mathbf{p}$.

For the second point, assume there are two destabilising representations contain $s(e)$ and $t(e)$ respectively, then either their sum or intersection is destabilising, which corresponds to the two cases discussed in the first point. $\square$

Remark. Without loss of generality, we may assume that all destabilising subrepresentations of $\mathbf{p} \setminus e$ contain $s(e)$. The set of such subrepresentations is finite and non-empty, denoted as $\mathcal{D}$.

Definition 2.1.18. Given an ordering on Q_1 , define the **incidence map** $\text{inc} : \mathbb{R}^{Q_1} \rightarrow \mathbb{R}^{Q_0} : \chi_e \mapsto \chi_{t(e)} - \chi_{s(e)}$, where $\chi_e := (0, \dots, 1, \dots, 0)$ the basis element of e , $\chi_{t(e)}, \chi_{s(e)}$ are similarly defined as the basis elements of vertices $s(e)$ and $t(e)$.

For $\mathbf{p}$ a θ -stable representation, and e non-loop such that $\mathbf{p} \setminus e$ is θ -unstable. Define new parameter

$$\theta' := \theta - \beta_e \text{inc}(\chi_e)$$

where $\beta_e := \max \{\theta(\mathbf{q} \setminus e) \mid \mathbf{q} \setminus e \in \mathcal{D}\}$, it exists since the set $\mathcal{D}$ is finite.

Proposition 2.1.19. *Let $\mathbf{p}$ be θ -stable, and e non-loop be such that $\mathbf{p} \setminus e$ θ -unstable. Then $\mathbf{p} \setminus e$ is strictly θ' -semistable, here 'strictly' means semistable condition could NOT be refined as stable.*

Moreover, $\mathbf{p} \setminus e$ is an extension of two θ' -stable representations $\mathbf{p}_1$ and $\mathbf{p}_2$.

Proof. By the construction of θ' , we know for those subrepresentations in $\mathcal{D}$, the maximality of β_e guarantees that they are all θ' -semistable. While for those θ -stable subrepresentations of $\mathbf{p} \setminus e$, if one of them is θ' -unstable, then its intersection or sum with a θ -maximising element is θ -positive, gives a θ -destabilising subrepresentation of $\mathbf{p}$, which is absurd. Thus all subrepresentations of $\mathbf{p} \setminus e$ would NOT destabilising $\mathbf{p} \setminus e$, moreover for the element $\mathbf{q} \setminus e$ in $\mathcal{D}$ that is θ -maximizing, we have $\theta'(\mathbf{q} \setminus e) = 0$, this implies that the representation $\mathbf{p} \setminus e$ is strictly θ' -semistable.

For the second point, if there are two elements in $\mathcal{D}$, say $\mathbf{q}_1$ and $\mathbf{q}_2$ being θ -maximising, then $\mathbf{q}_1 \cap \mathbf{q}_2$ and $\mathbf{q}_1 + \mathbf{q}_2$ will be in $\mathcal{D}$, and $\theta(\mathbf{q}_1 \cap \mathbf{q}_2) + \theta(\mathbf{q}_1 + \mathbf{q}_2) = \theta(\mathbf{q}_1) + \theta(\mathbf{q}_2)$. By maximality of β_e we have,

$$\theta(\mathbf{q}_1 \cap \mathbf{q}_2) = \theta(\mathbf{q}_1 + \mathbf{q}_2) = \beta_e$$

Since θ is generic, then either $\mathbf{q}_1 + \mathbf{q}_2$ equals to $\mathbf{q}_1 \cap \mathbf{q}_2$, or $\mathbf{q}_1$ is complement to $\mathbf{q}_2$. It is impossible since $\mathbf{q}_1, \mathbf{q}_2$ are non-zero, different subrepresentations of $\mathbf{p} \setminus e$. Hence there is a unique θ -maximising element in $\mathcal{D}$, denoted by $\mathbf{p}_1$, for any proper subrepresentation $\mathbf{p}'_1$ of $\mathbf{p}_1$, $\mathbf{p}'_1$ is also a destabilising representation for $\mathbf{p} \setminus e$, thus in $\mathcal{D}$, by uniqueness of $\mathbf{p}_1$, we have $\theta'(\mathbf{p}'_1) < 0$ being strictly negative, hence $\mathbf{p}_1$ is θ' -stable. Take $\mathbf{p}_2$ to be the quotient of $\mathbf{p} \setminus e$ by $\mathbf{p}_1$, i.e. $\mathbf{p}_2 = (\mathbf{p} \setminus e) / \mathbf{p}_1$. $\square$

With Proposition 2.1.19, we could provide an algorithm to give each $\mathbf{p} \in \mathcal{R}(Q^\#)$ a corresponding spanning tree T .

- If an edge e such that $\mathbf{p} \setminus e$ is θ -stable, then we move to $\mathbf{p} \setminus e$.
- Otherwise $\mathbf{p} \setminus e$ is θ -unstable, we take e to be one of the edges in the tree T we want. Further, by Proposition 2.1.19, it also provides two θ' -stable representations $\mathbf{p}_1$ and $\mathbf{p}_2$.
- Repeat above steps for $\mathbf{p}_1, \mathbf{p}_2$ (with respect to new parameter θ'), connect two subtrees by e , e goes from $\mathbf{p}_2$ to $\mathbf{p}_1$.
- The algorithm ends when there is only one vertex left.

Remark. With the algorithm, $\mathbf{p}$ can be associated to a unique spanning tree T .

Notation. We denote by $\mathcal{R}(Q^\#)_T \subseteq \mathcal{R}(Q^\#)$ the space contains all points associated to a fixed spanning tree T .

We denote by $Z_{\lambda,T}^\theta$ the intersection $Z_\lambda \cap \mathcal{R}(Q^\#)_T^\theta$, where $\mathcal{R}(Q^\#)^\theta$ denote for θ -stable points.

Lemma 2.1.20. *Use the notations above, for each $\mathbf{p} \in \mathcal{R}(Q^\#)_T$, we have $T^\theta \in K_{\mathbf{p}}^o$.*

Notation. Let $\mathcal{M}$ denote the Nakajima quiver variety $\mathcal{M}_{\lambda,\theta}(\mathbf{v})$. In our toric case, $\mathbf{v} = (1, \dots, 1)$.

Let $\mathcal{M}/e$ denote the contracted quiver variety by e with respect to contracted parameters λ/e and θ/e and dimension vector $\mathbf{v} = (1, \dots, 1)$ of length reduced by 1. Let $\mathcal{M} \setminus e$ denote the deleted quiver variety.

Let $\mathcal{M}_T$ denote the GIT quotient of $Z_{\lambda,T}^\theta$ by $\mathbf{G}_{\mathbf{v}}$.

We are ready to study the cell decomposition for the toric quiver variety $\mathcal{M}$.

Theorem 2.1.21. *Let T be a spanning tree of Q and let e be the biggest non-loop edge in Q . If $e \in T$ then*

$$\mathcal{M}_T \simeq (\mathcal{M}/e)_{T/e}$$

where $\mathcal{M}/e$ is the quiver variety on the contracted quiver $\tilde{Q}/e$ with toric dimension vector $\mathbf{v} = (1, \dots, 1)$ of length lower by 1 and contracted hyperkähler parameters $\lambda/e, \theta/e$.

Proof. Without loss of generality, we may assume that $T^\theta = T$, since $e \in T = T^\theta$, by Lemma 2.1.20 and the definition of $K_{\mathbf{p}}^o$, we know that $\mathbf{x}_e$ is an isomorphism. Hence the map $\mathbf{p} \mapsto \mathbf{p}/e$ gives a morphism $\mathcal{M}_T \rightarrow \mathcal{M}/e$ as $\mathbf{p}/e$ is θ/e -stable. To show the image of this morphism lies in $(\mathcal{M}/e)_{T/e}$ is equivalent to show $\mathbf{p}/e \in \mathcal{R}((Q/e)^\#)_{T/e}$. We do the algorithm on $\mathbf{p}/e$.

If we are in the first case of the algorithm, e' the smallest edge such that $\mathbf{p} \setminus e'$ is θ -stable, then $(\mathbf{p}/e) \setminus e'$ is again θ/e -stable. If e' is NOT stable, then since $e \in T$, then two endpoints $s(e), t(e)$ of e belong to $\mathbf{p}_1$ or $\mathbf{p}_2$. Recall the construction of $\mathbf{p}_1$ in the Proposition 2.1.19, $\mathbf{p}_1$ is a θ -maximizing destabilising representation of $\mathbf{p} \setminus e'$, then $\mathbf{p}_1/e$ is a destabilising representation of $(\mathbf{p}/e) \setminus e$. Hence the algorithm for $\mathbf{p}/e$ provides the same tree as T/e .

And conversely, point in $(\mathcal{M}/e)_{T/e}$ can be lifted to $\mathcal{M}$, we first need to give a precise construction of the lift, then show that the associated tree to the lifted point is exactly the tree T . We take $\mathbf{q} \in (Z/e)_{\lambda/e, T/e}^{\theta/e}$.

To construct the linear operators for each edge in $Q^\#$ from $\mathbf{q}$, we first let those linear operators associated to edges that not incident to $s(e)$ or $t(e)$ be unchanged. Since we have a priori isomorphisms $\phi : V_{s(e)} \rightarrow V_t$ and $\psi : V_t \rightarrow V_{t(e)}$, then take $\mathbf{x}_e$ to be $\psi \circ \phi$. Note here different choice of ϕ and ψ may lead to ~~same~~ $\mathbf{x}_e$, but after quotient by $\mathbf{G}_{\mathbf{v}}$ -action, they turns out to be the same case. Moreover, for those edges incident to one of $s(e)$ or $t(e)$, their corresponding linear maps could be composition with $\mathbf{x}_e$. And the only ambiguity here is $\mathbf{x}_e^*$, since all other linear operators are determined, we can use the moment map $\sum_{h \in Q_1} [\mathbf{x}_h, \mathbf{x}_h^*]$ to determine it. We get a point $\mathbf{p} \in Z_\lambda$, since $\mathbf{x}_e$ is an isomorphism by construction, we know $T \subseteq K_{\mathbf{p}}^o$.

To check that $\mathbf{p}$ constructed above is in Z_T , need to show the associated tree to $\mathbf{p}$ is exactly T . Suppose NOT, there is $T' \neq T$ such that T' associated to $\mathbf{p}$. That is to say, after running the algorithm for $\mathbf{p}$, we get T' ; for $\mathbf{q}$, we get T/e .

Let e' be the smallest non-loop edge, there are three possibilities:

- (i) If $e' \notin T'$, $e' \in T$, then $\mathbf{p} \setminus e'$ is stable. $\mathbf{x}_e$ is an isomorphism, and thus $(\mathbf{p} \setminus e')/e$ is also stable. Hence $e' \notin T$, a contradiction to the assumption.
- (ii) If $e' \notin T, e' \in T'$, then $T \subseteq K_{\mathbf{p} \setminus e'}^o$, since $e' \notin T$ is stable. By Proposition 2.1.16, we know $\mathbf{p} \setminus e'$ is stable. Since T' is the associated tree of $\mathbf{p}$, $e' \notin T'$, a contradiction to the assumption $e' \in T'$.
- (iii) If $e' \in T, e' \in T'$, the decomposition of $\mathbf{p}$ into $\mathbf{p}_1$ and $\mathbf{p}_2$ is different from the corresponding decomposition of $\mathbf{q}$ into $\mathbf{q}_1$ and $\mathbf{q}_2$. This can only happen if e connects $\mathbf{p}_1$ to $\mathbf{p}_2$ so that after contraction of e , $\mathbf{p}_1$ fails to be a subrepresentation of $\mathbf{q}$. By the maximality of $\mathbf{p}_1$ and $\mathbf{q}_1$ from their construction, we are forced to take $\theta(\mathbf{p}_1) > \theta(\mathbf{q}_1)$ so that contradicts to the maximality of $\mathbf{q}_1$.

Set all maps of $\mathbf{p}$ which are not in T to zero. Denote the resulting representation by $\mathbf{p}'$ and the subrepresentation corresponding to $\mathbf{p}_1$ by $\mathbf{p}'_1$. Apply the algorithm to $\mathbf{p}'$, this produces a tree contained in T , so it must be T . In particular, the first step of the algorithm produces a subrepresentation which goes to $\mathbf{q}_1$ after contraction by e , so we have $\theta(\mathbf{q}_1) \geq \theta(\mathbf{p}'_1) = \theta(\mathbf{p}_1)$, a contradiction to the previous fact that $\theta(\mathbf{p}_1) > \theta(\mathbf{q}_1)$. Hence $\mathbf{q}$ lift to $\mathbf{p}$ with the associated tree to be exactly T . $\square$

Theorem 2.1.22. *Let T be a spanning tree of Q and e be the biggest non-loop edge in Q . If $e \notin T$ then*

$$\mathcal{M}_T \simeq (\mathcal{M} \setminus e)_T \times \mathbb{A}_{\mathbf{k}}^1$$

where $(\mathcal{M} \setminus e)_T$ is the quiver variety on the deleted quiver $\vec{Q} \setminus e$ with dimension vector and hyperkähler parameters unchanged.

The idea of proof is similar to the previous one, before proving the theorem, we introduce the following conception.

Consider T being a spanning tree of Q , then adding any extra edge on T will produce a cycle. Assume $e \notin T$, we denote by $C(T, e)$ the cycle in Q obtained by adding e on T , note that it is unique if we neglect the orientation. For $e \neq h \in C(T, e)$, since $h \in T$, the orientation of h in T^θ gives rise to direction of the circle, thus an orientation of e , called the **h -induced orientation**. Note that for two edges $e \neq h_1, h_2 \in C(T, e)$, h_1, h_2 -induced orientations might be reversed.

Lemma 2.1.23. *Let $\theta \in \mathbb{R}^{Q_0}$, T be a spanning tree of Q , e be the biggest non-loop edge in Q and $\mathbf{p} = (\mathbf{x}, \mathbf{x}^*) \in \mathbf{Z}_{\lambda, T}$. Take $e \notin T$ and let h be the smallest edge in $C(T, e)$. Assume that the h -induced orientation on e is opposite to its orientation as an edge in Q_1 , then $\mathbf{x}_e = 0$, otherwise if the orientation is the same, $\mathbf{x}_e^* = 0$.*

Proof. Since $\mathbf{p} \in \mathcal{M}_T$ and $e \notin T$, there exist h the smallest edge such that $\mathbf{p} \setminus h$ is θ -unstable. Assume that the h -induced orientation on e is opposite to its orientation. If $\mathbf{x}_e \neq 0$, we can consider the spanning tree $T' := (T \setminus h) \cup e$ such that $T'^\theta \subseteq K_{\mathbf{p}}^o$, a contradiction to the assumption. $\square$

Proof of Theorem 2.1.22. Let $\mathbf{q} \in (\mathbf{Z} \setminus e)_{\lambda, T}$ the point of representation of deleted quiver, to give a lift to $\mathbf{p} \in \mathbf{Z}_{\lambda, T}$, we first let those linear operators associated to edges not equal to e be unchanged. With Lemma 2.1.23, we could assume that $\mathbf{x}_e = 0$ by choosing a proper orientation, while $\mathbf{x}_e^* \in \text{Hom}(V_{t(e)}, V_{s(e)})$ could be chosen arbitrarily.

Under such construction, the θ -stability comes from Proposition 2.1.16, and since $\mathbf{x}_e = 0$, the value of the moment map on $\mathbf{p}$ is same as $\mathbf{q}$. We apply similarly the method in Proof of Theorem 2.1.21, the spanning tree of $\mathbf{p}$ is T , hence $\mathbf{p} \in \mathbf{Z}_{\lambda}^\theta$.

Conversely, for $\mathbf{p} \in \mathcal{M}_T$, deleting the edge e provides naturally a representation $\mathbf{q}$ of $\vec{Q} \setminus e$. Then from above discussion, $\mathbf{q}$ can be recovered to original representation $\mathbf{p}$ with same tree associated. $\square$

Theorem 2.1.24. *For a spanning tree $T \subseteq Q$, we have*

$$\mathcal{M}_T \simeq \mathbb{A}_{\mathbf{k}}^{b_1(Q) + \text{ext}(T)}$$

In the formula, the function b_1 denotes the first Betti number of a graph, given by $b_1(\Gamma) = |E| + |C| - |V|$, where E : set of edges, C : set of connected components, V : set of vertices.

Proof. For a connected quiver Q , and $T \subseteq Q$ a spanning tree, since deletion will NOT glue vertices, the number of times of contractions to count external activity of T is $\#Q_0 - 1$, while there is $\text{ext}(T)$ edges left at last, the number of times of deletions is $\#Q_1 - (\#Q_0 - 1) - \text{ext}(T) = b_1(Q) - \text{ext}(T)$.

Consider the case when $\vec{Q}$ is the quiver of one vertex and several loops, and the corresponding toric quiver variety $\mathcal{M}_{\lambda, \theta}(\vec{Q}, \mathbf{v})$. By genericity of λ , it must be zero as $\lambda \cdot \mathbf{v} = \lambda = 0$. And the choice of linear operators in $\text{Hom}_{\mathbf{k}}(\mathbf{k}, \mathbf{k}) \oplus \text{Hom}_{\mathbf{k}}(\mathbf{k}, \mathbf{k})$ is arbitrary, each choice will give an isomorphism class of indecomposable representations, and thus isomorphism class of completely indecomposable representations. Hence $\mathcal{M} := \mathcal{M}_{0,0}(\vec{Q}, 1) = \mathbb{A}_{\mathbf{k}}^{2(\#Q_1)}$, $|\mathcal{M}(\mathbb{F}_q)| = q^{2(\#Q_1)}$. Combining the Theorem 2.1.21 and Theorem 2.1.22 with the counting of number of contractions/deletions, we know for a spanning tree $T \subseteq Q$,

$$\mathcal{M}_T \simeq \mathbb{A}_{\mathbf{k}}^{2\text{ext } T} \times \mathbb{A}_{\mathbf{k}}^{b_1(Q) - \text{ext}(T)} \simeq \mathbb{A}_{\mathbf{k}}^{b_1(Q) + \text{ext}(T)}.$$

This gives a cell decomposition of toric quiver varieties by spanning trees. $\square$

Corollary 2.1.25. *The Poincaré polynomial of $\mathcal{M}$ is given by*

$$P_{\mathcal{M}}(q) = q^{b_1(Q)} \cdot \mathbf{T}_Q(1, q)$$

Chapter 3

More on Kac's Conjecture

In the following sections, we would like to give an overview of Kac's conjecture, including its background and connection with quiver varieties. We start with introducing Kac's paper.

3.1 Kac-Moody Lie Algebras and Representations of Graphs

Due to the work of Gabriel, we know that for a Dynkin quiver $\vec{Q}$, there is a bijection between the set of isomorphism classes of indecomposable representations and the positive roots of the corresponding finite-dimensional simple Lie algebras. And the following works gave the board between so called 'tame' and 'wild' quivers, and showed that all tame quivers have underlying graphs of Euclidean type.

We describe here a brief introduction to Kac-Moody Lie algebras and relations to representations of graphs.

Definition 3.1.1. A matrix $C = (c_{ij})_{i,j=1,\dots,n}$ of integer coefficient is called a **Cartan matrix** if the following conditions hold:

(C1) $c_{ii} = 2$;

(C2) $-c_{ij} \in \mathbb{N}$ for $i \neq j$;

(C3) $c_{ij} = 0$ implies $c_{ji} = 0$ for $i, j = 1, \dots, n$.

A matrix $A = (a_{ij})$ is said to be **symmetrisable** if the following conditions hold:

(S1) $a_{ij} = 0$ implies $a_{ji} = 0$;

(S2) $a_{i_1 i_2} a_{i_2 i_3} \dots a_{i_k i_1} = a_{i_1 i_k} a_{i_k i_{k-1}} \dots a_{i_2 i_1}$ for any indices set $i_1, \dots, i_k$.

A matrix is **decomposable** if it is equivalent (up to conjugation by permutation matrix) to a matrix of block diagonal form.

We have the following evidence:

Lemma 3.1.2. (i) The matrix A is symmetrisable if and only if it can be represented in the form $A = D \cdot B$, where D is a non-degenerate diagonal matrix and B is a symmetric matrix.

(ii) If C is an indecomposable symmetrisable Cartan matrix, then there exists a unique decomposition

$$C = D \cdot B = \text{diag}(d_1, \dots, d_n) \cdot (b_{ij})$$

such that $b_{ij} = b_{ji}$ are non-positive half-integers for $i \neq j$, $b_{ii} = 2d_i^{-1}$ are relatively prime positive integers.

Lemma 3.1.3. Assume C is an indecomposable Cartan matrix, then C is of one and only one of the following cases:

(P) C is of full rank, i.e. $\det C \neq 0$, and $Cu \geq 0$ implies $u \geq 0$ or $u = 0$.

(Z) C is degenerate and $\text{corank}(C) = 1$; there exists $u > 0$ such that $Cu = 0$, and $Cv \geq 0$ implies $Cv = 0$.

(N) There exists $u > 0$ such that $Cu < 0$; $v \geq 0$ and $Cv \geq 0$ implies $v = 0$.

The Cartan matrices in above cases are called of **positive, zero and negative** type respectively. (Also known as **finite, affine and indefinite** type.)

Remark. In the cases of positive and zero type, the Cartan matrix C is symmetrisable and thus we can apply Lemma 3.1.2 to give C a diagonal-symmetric decomposition. Moreover, the symmetric matrix B in the decomposition is positive-definite or positive semi-definite respectively.

Definition 3.1.4. Let $C = (c_{ij})_{i,j \in I}$ be a Cartan matrix, where I is a finite index set. Then we can associate a graph to C , called the **valued Dynkin diagram** $S(C)$ associated to C . The vertices set of $S(C)$ is I , denote each vertex by $p_i, i \in I$. The vertices p_i, p_j are joined by $k_{ij} = \min(|c_{ij}|, |c_{ji}|)$ edges and with a ordered pair $(|c_{ji}|, |c_{ij}|)$ on edges connecting p_i and p_j . Conversely, given a graph S , we could also associate a generalised Cartan matrix C .

We have also associated (complex) Lie algebra with root system and the Weyl group arising from the Cartan matrix.

Definition 3.1.5. Choose C to be a $(n \times n)$ Cartan matrix, and let Γ be a free abelian group with generators $\alpha_1, \dots, \alpha_n$. The associated Lie algebra to C is a Γ -graded one $\mathfrak{g}(C) := \bigoplus_{\alpha \in \Gamma} \mathfrak{g}_\alpha$ satisfying the following:

(i) Every graded ideal which intersects $\mathfrak{g}_0$ trivially is zero; (ii) $\mathfrak{g}(C)$ is generated by elements e_i, f_i, h_i such that $\mathfrak{g}_{\alpha_i} = \mathbb{C}e_i, \mathfrak{g}_{-\alpha_i} = \mathbb{C}f_i, h_i$'s form a basis of $\mathfrak{g}_0$, and the following **Serre relations** hold:

$$\begin{aligned} [h_i, h_j] &= 0, & [e_i, f_j] &= \delta_{ij} h_i, & [h_i, e_j] &= c_{ij} e_j, & [h_i, f_j] &= -c_{ij} f_j \\ \text{ad}(e_i)^{1-c_{ij}} e_j &= 0, & \text{ad}(f_i)^{1-c_{ij}} f_j &= 0. \end{aligned}$$

$\mathfrak{g}(C)$ is called a **Kac-Moody Lie algebra**.

Definition 3.1.6. An element $\alpha \in \Gamma$ is called a **root** of the Lie algebra $\mathfrak{g}(C)$ if $\mathfrak{g}_\alpha \neq 0$; the number $\dim \mathfrak{g}_\alpha$ is called the **multiplicity** of the root α . We denote by $\Delta = \Delta(C)$ the set of all roots.

And the roots $\alpha_1, \dots, \alpha_n$ are called **simple**, we denote by $\Pi = \{\alpha_1, \dots, \alpha_n\}$ the set of simple roots.

Let Γ_+ be the set of all non-zero elements in Γ of the form $\alpha = k_1 \alpha_1 + \dots + k_n \alpha_n$ with $k_i \geq 0, i = 1, \dots, n$; If α is a root, then $-\alpha$ is also a root of the same multiplicity and one of these roots belongs to Γ_+ . The elements from $\Delta_+ = \Gamma_+ \cap \Delta$ are called **positive roots**. Those roots α such that $-\alpha \in \Delta_+$ are called **negative**.

We set $L = \Gamma \otimes_{\mathbb{Z}} \mathbb{R}$ and introduce linear functions $\varphi_1, \dots, \varphi_n$ on L by setting $\varphi_i(\alpha_j) = c_{ij}$. The induced endomorphisms r_i on L (called **reflections**) is given by

$$r_i(\alpha) := \alpha - \varphi_i(\alpha) \alpha_i, \alpha \in L$$

Definition 3.1.7. The subgroup $W = W(C)$ of $\text{GL}(L)$ generated by $r_1, \dots, r_n$ is called the **Weyl group** (of $\mathfrak{g}(C)$).

Remark. The Weyl group defined above is a Coxeter group, with relations of reflections are given by

$$r_i^2 = 1; \quad (r_i r_j)^{n_{ij}} = 1,$$

where n_{ij} 's are obtained from the table:

$c_{ij} c_{ji}$	0	1	2	3	≥ 4
n_{ij}	2	3	4	6	∞

Definition 3.1.8. We say a root $\alpha \in \Delta$ is **real** if $w(\alpha)$ is simple for some $w \in W$, all other roots not real are called **imaginary**. We denote by $\Delta^{\text{re}}, \Delta_+^{\text{re}}, \Delta^{\text{im}},$ and Δ_+^{im} the sets of all real, positive real, imaginary and positive imaginary roots respectively.

Consider if the Cartan matrix C is decomposable, assume $C_1, \dots, C_r$ are its diagonal blocks, then we have $\mathfrak{g}(C)$ be a direct sum of $\mathfrak{g}(C_i)$, the root space $\Delta(C)$ is the disjoint union of $\Delta(C_i)$ and the Weyl group $W(C)$ is the product of $W(C_i)$ while each component $W(C_i)$ acts on $\Delta(C_j)$ trivially for $i \neq j$. Hence we may consider from now on C to be indecomposable.

Now assume that C is a $(n \times n)$ symmetrisable Cartan matrix, with diagonal-symmetric decomposition $C = D \cdot B$. Define a symmetric bilinear form $(\cdot, \cdot)$ on Γ by setting $(\alpha_i, \alpha_j) = b_{ij} = b_{ji} = (\alpha_j, \alpha_i)$.

Lemma 3.1.9. We have the bilinear form $(\cdot, \cdot)$ defined above to be W -invariant, and

- (i) A root α is real if and only if $(\alpha, \alpha) > 0$.
- (ii) A root α is imaginary if and only if $(\alpha, \alpha) \leq 0$.

Theorem 3.1.10 (Kac 1980 [Kac80]). Let $\vec{Q}$ be a quiver with Q the underlying graph and assume that the field $\mathbf{k}$ is algebraically closed.

For any $\alpha \in \Gamma_+ = \sum_{i=1}^n k_i \alpha_i$, we set the corresponding dimension vector $\mathbf{v}_\alpha = (k_1, \dots, k_n)$, then

- (i) For $\alpha \in \Gamma_+ \setminus \Delta_+$, any representation of $\vec{Q}$ of dimension $\mathbf{v}_\alpha$ over $\mathbf{k}$ is decomposable.

- (ii) For $\alpha \in \Delta_+$, there exists an indecomposable representation of $\vec{Q}$ of dimension $\mathbf{v}_\alpha$. Moreover, the measure of set of indecomposable representations of dimension $\mathbf{v}_\alpha$ is $\geq 1 - (\alpha, \alpha)$.
- (iii) For $\alpha \in \Delta_+^{\text{re}}$, there exists one and only one indecomposable representation of dimension $\mathbf{v}_\alpha$.

Remark. Note that for positive (finite) type Kac-Moody algebra, all roots are real, hence the above theorem gives a generalization of Gabriel's theorem.

Now let $\mathbf{k} = \mathbb{F}_q$ be the finite field, for q is prime power. A representation $(V, \mathbf{x})$ of $\vec{Q}$ over $\mathbf{k}$ is absolutely indecomposable if $V \otimes_{\mathbf{k}} \overline{\mathbb{F}_q}$ is indecomposable over $\overline{\mathbb{F}_q}$.

For any $\alpha \in \Delta_+$, we let $A_Q(\alpha, q)$ denote the number of isomorphism classes of absolutely indecomposable representation of $\vec{Q}$ over $\mathbf{k}$ with dimension $\mathbf{v}_\alpha$. In [Kac80], Kac showed that $A_Q(\alpha, q)$ is a polynomial in q of integer coefficients and is independent on the orientation of $\vec{Q}$. Kac also made the following conjecture:

- Conjecture 3.1.11** (Kac). (i) The coefficients of $A_Q(\alpha, q)$ are always non-negative.
- (ii) The constant term of $A_Q(\alpha, q)$ is m_α , the multiplicity of root α in $\mathfrak{g}$.

3.2 Modules of Kac-Moody Lie algebras

By our construction of Kac-Moody algebra $\mathfrak{g}(C)$, we know that there is a root decomposition of $\mathfrak{g}$ and its universal enveloping algebra $\mathcal{U}(\mathfrak{g})$.

$$\mathfrak{g} = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+; \quad \mathcal{U}(\mathfrak{g}) = \mathcal{U}(\mathfrak{n}_-) \otimes \mathcal{U}(\mathfrak{h}) \otimes \mathcal{U}(\mathfrak{n}_+) = \mathcal{U}(\mathfrak{n}_-) \otimes \mathcal{U}(\mathfrak{b}).$$

where $\mathfrak{h} = \mathfrak{g}_0$, $\mathfrak{n}_+ = \bigoplus_{\alpha \in \Delta_+} \mathfrak{g}_\alpha$ and $\mathfrak{n}_- = \bigoplus_{\alpha \in \Delta_+} \mathfrak{g}_{-\alpha}$. Hence simple roots $\alpha_i \in \mathfrak{h}^*$ and for h_i the basis of $\mathfrak{h}$, we call them simple coroots, denote by $\Pi^\vee = \{h_1, \dots, h_n\} = \{\alpha_i^\vee\}$.

Choose $\mathfrak{h}_{\mathbb{R}}$ such that the complexification of $\mathfrak{h}_{\mathbb{R}}$ is $\mathfrak{h}$. With respect to the symmetric bilinear form $(\cdot, \cdot)$, we introduce

$$\langle \cdot, \cdot \rangle : \mathfrak{h} \times \mathfrak{h}^* \rightarrow \mathbb{Z}$$

satisfying: $\langle h_i, \alpha \rangle = (\alpha_i, \alpha)$ for $\alpha \in \Delta \subseteq \Gamma \subseteq \mathfrak{h}_{\mathbb{R}}^*$. Choose elements $\Lambda_i \in \mathfrak{h}_{\mathbb{R}}^*, i = 1, \dots, n$ such that $\langle \Lambda_i, h_j \rangle = \delta_{ij}$.

Definition 3.2.1. We call elements Λ_i **fundamental weights**, and let $P \subseteq \mathfrak{h}_{\mathbb{R}}^*$ be the free $\mathbb{Z}$ -module generated by Λ_i and α_i 's. We call P the **weight lattice**.

Definition 3.2.2. Let M (not necessarily finite dimensional) be a $\mathcal{U}(\mathfrak{g}(C))$ -module. A vector $v \in M$ is said to have **weight** $\lambda \in \mathfrak{h}^*$ if $h.v = \langle h, \lambda \rangle v$ for any $h \in \mathfrak{h}$. Denote $M_\lambda \subseteq M$ the subspace of vectors of weight λ .

We say that a $\mathcal{U}(\mathfrak{g}(C))$ -module M has a **weight decomposition** if for every $\lambda \in P$, $M = \bigoplus_{\lambda \in P} M_\lambda$. The set

$$P(M) = \{\lambda \in P \mid V_\lambda \neq 0\}$$

is called the **set of weights** of M .

We set for $\lambda \in \mathfrak{h}^*$ the set $D(\lambda) := \{\mu \in \mathfrak{h}^* \mid \mu \leq \lambda\}$. Fix $\mathfrak{g}(C)$ a Kac-Moody Lie algebra.

Definition 3.2.3. We define the category $\mathcal{O}$ to be the full subcategory of the category of $\mathcal{U}(\mathfrak{g}(C))$ -modules whose objects are $\mathcal{U}(\mathfrak{g})$ -modules M which have weight decompositions with each weight space M_λ being finite dimensional and there exist elements $(\lambda_i \in \mathfrak{h}^*)_{i=1, \dots, s}$ such that

$$P(M) \subset \bigcup_{i=1}^s D(\lambda_i)$$

Call this category **BGG category $\mathcal{O}$** .

Remark. The submodules and quotients of objects in $\mathcal{O}$ are still in $\mathcal{O}$. Finite direct sums and finite tensors of objects in $\mathcal{O}$ are still in $\mathcal{O}$.

Definition 3.2.4. Assume that $M \in \mathcal{O}$ is a $\mathcal{U}(\mathfrak{g})$ -module. A nonzero vector v_λ is said to be a **maximal vector** of weight λ if $v_\lambda \in M_\lambda$ and $\mathfrak{n}_+.v = 0$. The corresponding weight λ is called a **primitive weight**.

And M is said to be a **highest weight module** of weight λ if there is a maximal vector $v_\lambda \in M_\lambda$ such that M is generated by v_λ , i.e. $M = \mathcal{U}(\mathfrak{g}).v_\lambda$. By the property of v_λ , we know $M \simeq \mathcal{U}(\mathfrak{n}_-).v_\lambda$.

Definition 3.2.5. For $\lambda \in \mathfrak{h}^*$, we set $\mathbb{C}_\lambda$ to be a one-dimensional $\mathfrak{b}$ -module, with the $\mathfrak{n}_+$ -action to be trivial. Then we consider the induced $\mathcal{U}(\mathfrak{g})$ -module $M(\lambda)$ given by $M(\lambda) = \text{Ind}_{\mathfrak{b}}^{\mathfrak{g}} \mathbb{C}_\lambda := \mathcal{U}(\mathfrak{g}) \otimes_{\mathcal{U}(\mathfrak{b})} \mathbb{C}_\lambda$, called the **Verma module**.

Since $\mathcal{U}(\mathfrak{g}) \simeq \mathcal{U}(\mathfrak{n}_-) \otimes \mathcal{U}(\mathfrak{b})$, $M(\lambda)$ is isomorphic to $\mathcal{U}(\mathfrak{n}_-) \otimes \mathbb{C}_\lambda$ as left $\mathcal{U}(\mathfrak{n}_-)$ -module, which is free of rank one. Let $v_\lambda := 1 \otimes 1$, then it is clear that $\mathfrak{n}_+.v_\lambda = 0$ and $h.v_\lambda = \langle \lambda, h \rangle v, \forall h \in \mathfrak{h}$. And since v_λ generates $M(\lambda)$, it is a maximal vector.

By PBW theorem, the left ideal I_λ of $\mathcal{U}(\mathfrak{g})$ which annihilates $v_\lambda = 1 \otimes 1$ is generated by $\mathfrak{n}_+$ and $h - \langle \lambda, h \rangle \cdot 1, \forall h \in \mathfrak{h}$. Hence $M(\lambda) \simeq \mathcal{U}(\mathfrak{g})/I_\lambda$. For M any other highest weight module of weight λ , I_λ also annihilates M . Thus M is as a quotient of $M(\lambda)$ and the Verma module is regarded as universal highest weight module.

Assume M is a highest weight module of weight λ , then each submodule of M has weight decomposition. Since M is generated by the one-dimensional subspace M_λ , any proper submodule of M could NOT have weight λ , hence sum of all proper submodule is still proper and M has a unique maximal submodule.

Denote by $L(\lambda) = M(\lambda)/N(\lambda)$ the unique irreducible quotient of $M(\lambda)$, where $N(\lambda)$ is the unique maximal submodule of $M(\lambda)$.

Proposition 3.2.6. These $L(\lambda)$ are all irreducible highest weight module in the BGG category $\mathcal{O}$.

Definition 3.2.7. A $\mathfrak{g}(C)$ -module M is called integrable if it is locally nilpotent, that is, for each $i \in \{1, \dots, n\}$ and $v \in M$, there exists N such that $e_i^N.v = f_i^N.v = 0$.

Theorem 3.2.8. An irreducible highest weight module $L(\lambda)$ is integrable if and only if $\lambda \in P_+$, where

$$P_+ = \{\lambda \in P \mid \langle \lambda, \alpha_i^\vee \rangle = \langle \lambda, h_i \rangle \in \mathbb{Z}_+, \forall i\}$$

those weights in P_+ are called **dominant weights**.

3.2.1 Character Formula

Definition 3.2.9. Let M be a module in the BGG category $\mathcal{O}$, then we define the **formal character** $\text{ch}(M)$ associate to M by setting

$$\text{ch } M = \sum_{\lambda \in \mathfrak{h}^*} (\dim M_\lambda) e(\lambda)$$

where $e(\lambda)$ is a symbol associate to λ , in suggestive of formal exponential, satisfying $e(\lambda)e(\mu) = e(\lambda + \mu)$.

Lemma 3.2.10 ([Kac90] Lemma 9.6). Let M be a $\mathfrak{g}(C)$ -module from the category $\mathcal{O}$ and let $\lambda \in \mathfrak{h}^*$. Then there exists a submodule filtration of M given by $0 = M_0 \subseteq M_1 \subseteq \dots \subseteq M_t = M$ and a subset $J \subseteq \{1, \dots, t\}$ such that:

- (i) If $j \in J$, then $M_j/M_{j-1} \simeq L(\lambda_j)$ for some $\lambda_j \geq \lambda$;
- (ii) If $j \notin J$, then $(M_j/M_{j-1})_\mu = 0$ for any $\mu \geq \lambda$.

Let $M \in \mathcal{O}$ and $\mu \in \mathfrak{h}^*$. Fix $\lambda \in \mathfrak{h}^*$ such that $\mu \geq \lambda$ and take the filtration as in Lemma 3.2.10. Then denote by $[M : L(\mu)]$ the number of times μ appears among $\{\lambda_j \mid j \in J\}$.

The number $[M : L(\mu)]$ is in fact independent of the filtration and of the choice of λ , and is called the **multiplicity** of $L(\mu)$ in M . Note that $L(\mu)$ has a non-zero multiplicity in M if and only if μ is a primitive weight of M .

Proposition 3.2.11. Let M be a $\mathfrak{g}(C)$ -module from the category $\mathcal{O}$, then we have

$$\text{ch } M = \sum_{\lambda \in \mathfrak{h}^*} [M : L(\lambda)] \text{ch } L(\lambda)$$

Proof. Let $\phi(M)$ be the difference $\text{ch}(M) - \sum_{\lambda \in \mathfrak{h}^*} [M : L(\lambda)] \text{ch } L(\lambda)$. And take the filtration as in Lemma 3.2.10.

It is clear to see that for an exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ in $\mathcal{O}$, the character formula satisfies: $\text{ch}(M) = \text{ch}(M') + \text{ch}(M'')$. Then $\phi(M) = \sum_{j \notin J} \phi(M_j) + \sum_{j \in J} \phi(L(\lambda_j))$ with respect to the filtration. We know from definition that $\phi(L(\lambda_j)) = 0$ and thus $\phi(M) = \sum_{j \notin J} \phi(M_j)$. While by the consequence of the Lemma, for the fixed $\lambda \in \mathfrak{h}^*$, $(M_j)_\mu = 0$, for $j \notin J, \mu \geq \lambda$. Hence the coefficient of $e(\lambda)$ in $\phi(M)$ is zero. Changing the weight and filtration, we know $\phi(M)$ should be zero and the equality holds. $\square$

Proposition 3.2.12. *The character formula of $M(\lambda)$ is*

$$\text{ch } M(\lambda) = \prod_{\alpha \in \Delta_+} (1 - e(-\alpha))^{-m_\alpha} e(\lambda)$$

Proof. Let $\beta_1, \beta_2, \dots$ be all positive roots of $\mathfrak{g}$, and let $e_{-\beta_s, i_s}$ denote the basis of $\mathfrak{g}_{-\beta_s}$, where $1 \leq i_s \leq m_{\beta_s}$. For v_λ a highest weight vector of $M(\lambda)$, we know $M(\lambda)$ is generated by $\mathcal{U}(\mathfrak{n}_-).v_\lambda$. Thanks to the PBW theorem, the vectors

$$e_{-\beta_1, 1}^{n_{1,1}} \dots e_{-\beta_1, m_1}^{n_{1, m_1}} e_{-\beta_2, 1}^{n_{2,1}} \dots e_{-\beta_2, m_2}^{n_{2, m_2}} \dots v_\lambda$$

form the basis of $M(\lambda)_\mu$, where $(n_{1,1} + \dots + n_{1, m_1} \beta_1) + \dots = \lambda - \mu$. As a consequence, we have

$$\text{ch } M(\lambda) = \prod_{\alpha \in \Delta_+} (1 + e(-\alpha) + e(-2\alpha) + \dots)^{m_\alpha} e(\lambda) = \prod_{\alpha \in \Delta_+} (1 - e(-\alpha))^{-m_\alpha} e(\lambda)$$

□

We introduce the important and fundamental result in the representation theory of Kac-Moody algebra:

Theorem 3.2.13 ([Kac90], Theorem 10.4). *Let C be a symmetrisable Cartan matrix and $\mathfrak{g}(C)$ the associated Kac-Moody algebra. Assume $L(\lambda)$ is an integrable, irreducible $\mathfrak{g}(C)$ -module with highest weight $\lambda \in P_+$, then we have*

$$\text{ch } L(\lambda) = \frac{\sum_{w \in W} \epsilon(w) e(w(\lambda + \rho) - \rho)}{\prod_{\alpha \in \Delta_+} (1 - e(-\alpha))^{m_\alpha}} \quad (3.1)$$

where $\epsilon(w)$ denotes for the determinant $\det_{\mathfrak{h}^*}(w)$ of w on the space $\mathfrak{h}^*$, and ρ in the formula is the solution of the system $\langle \rho, \alpha_i^\vee \rangle = 1, i = 1, \dots, n$. The formula is known as **Kac-Weyl formula**.

Remark. (i) Note that ρ is unique in finite type case, which equals to half of the sum over positive roots, note also $\epsilon(w) = (-1)^{\ell(w)}$ can be computed by the length of $w \in W$.

(ii) Take $\lambda = 0$ in the formula 3.1, since $L(0)$ is trivial one-dimensional module, we have the following identity:

$$\prod_{\alpha \in \Delta_+} (1 - e(-\alpha))^{m_\alpha} = \epsilon(w) e(w(\rho) - \rho)$$

This is called **denominator identity**.

3.3 Geometric Realization of Kac-Moody Lie algebras

We continue the discussion of Nakajima quiver varieties to obtain a geometric interpretation of Kac-Moody algebra.

Fix a quiver $\vec{Q} = (Q_0, Q_1)$ with underlying graph Q . For the dimension vector $\mathbf{v}$, the framing vector $\mathbf{w}$, we fix a character χ of $G_\mathbf{v}$ which is determined by parameter θ . Recall that Nakajima's quiver varieties are given as:

$$\mathcal{M}_0(\mathbf{v}, \mathbf{w}) := \mu^{-1}(0) // G = \text{Spec } A(\mu^{-1}(0))^G; \quad \mathcal{M}(\mathbf{v}, \mathbf{w}) := \mu^{-1}(0) //_\chi G = \text{Proj } \bigoplus_{n \geq 0} A(\mu^{-1}(0))^G \cdot \chi^n.$$

In convenience, we abbreviate them as $\mathcal{M}_0$ and $\mathcal{M}$ respectively if no ambiguity.

Definition 3.3.1. For a point $(\mathbf{p}, \mathbf{i}, \mathbf{j})$ in the representation space $\mathcal{R}((Q^\vee)^\#, \mathbf{v}, \mathbf{w})$, we say it is **regular** if its orbit is closed and its stabilizer is trivial. Let $\mathcal{M}_0^{\text{reg}} \subseteq \mathcal{M}_0$ denote the regular locus of regular points.

Note there is possibility that $\mathcal{M}_0^{\text{reg}}$ being empty. We have the criterion for the non-emptiness:

Theorem 3.3.2. *Let θ be a $\mathbf{v}$ -generic parameter, and assume that $\mathcal{M}_0^{\text{reg}}$ is non-empty, then we have for any point $[\mathbf{p}, \mathbf{i}, \mathbf{j}] \in \mathcal{M}_0^{\text{reg}}$, it is stable and the natural projection: $\pi : \mathcal{M} \rightarrow \mathcal{M}_0$ is a symplectic resolution of singularities.*

For any dimension vectors $\mathbf{v}^1, \mathbf{v}^2 \in \mathbb{Z}_+^{Q_0}$, consider the map

$$\mathcal{R}(\mathbf{v}^1, \mathbf{w}) \rightarrow \mathcal{R}(\mathbf{v}^1 + \mathbf{v}^2, \mathbf{w}), \quad (\mathbf{p}, \mathbf{i}, \mathbf{j}) \mapsto (\mathbf{p} \oplus 0, \mathbf{i} \oplus 0, \mathbf{j} \oplus 0)$$

it is a $G_{\mathbf{v}^1}$ -equivariant map, hence the corresponding morphism $\mathcal{M}_0(\mathbf{v}^1, \mathbf{w}) \rightarrow \mathcal{M}_0(\mathbf{v}^1 + \mathbf{v}^2, \mathbf{w})$. The morphism is a closed embedding, and we can consider its composition with the projection morphism $\mathcal{M}(\mathbf{v}^1, \mathbf{w}) \rightarrow \mathcal{M}_0(\mathbf{v}^1, \mathbf{w})$, denote the composed morphism by $\pi_{\mathbf{v}^1}$. Define a variety $Z(\mathbf{v}^1, \mathbf{v}^2, \mathbf{w})$ to be the pull back of $\pi_{\mathbf{v}^1}, \pi_{\mathbf{v}^2}$.

$$\begin{array}{ccc} Z(\mathbf{v}^1, \mathbf{v}^2, \mathbf{w}) & \xrightarrow{\quad} & \mathcal{M}(\mathbf{v}^2, \mathbf{w}) \\ \downarrow & \searrow \pi_Z & \downarrow \pi_{\mathbf{v}^2} \\ \mathcal{M}(\mathbf{v}^1, \mathbf{w}) & \xrightarrow{\pi_{\mathbf{v}^1}} & \mathcal{M}(\mathbf{v}^1 + \mathbf{v}^2, \mathbf{w}) \end{array}$$

Then $Z(\mathbf{v}^1, \mathbf{v}^2, \mathbf{w}) = \mathcal{M}(\mathbf{v}^1, \mathbf{w}) \times_{\mathcal{M}(\mathbf{v}^1 + \mathbf{v}^2, \mathbf{w})} \mathcal{M}(\mathbf{v}^2, \mathbf{w})$ is a subvariety of $\mathcal{M}(\mathbf{v}^1, \mathbf{w}) \times \mathcal{M}(\mathbf{v}^2, \mathbf{w})$.

Definition 3.3.3. The variety $Z(\mathbf{v}^1, \mathbf{v}^2, \mathbf{w})$ is called a **steinberg variety**. Define also the regular elements in $Z(\mathbf{v}^1, \mathbf{v}^2, \mathbf{w})$ to be

$$Z^{\text{reg}} = \text{Int} \left(\left\{ x \in Z(\mathbf{v}^1, \mathbf{v}^2, \mathbf{w}) \mid \pi_Z(x) \in \mathcal{M}_0^{\text{reg}}(\mathbf{v}, \mathbf{w}) \text{ for some } \mathbf{v} \leq \mathbf{v}^1 + \mathbf{v}^2 \right\} \right)$$

here Int stands for the interior with respect to Zariski topology, and π_Z occurs in defining diagram of $Z(\mathbf{v}^1, \mathbf{v}^2, \mathbf{w})$.

Theorem 3.3.4. (i) For any irreducible component X of $Z(\mathbf{v}^1, \mathbf{v}^2, \mathbf{w})$, we have

$$\dim X \leq \frac{1}{2} \left(\dim \mathcal{M}(\mathbf{v}^1, \mathbf{w}) + \dim \mathcal{M}(\mathbf{v}^2, \mathbf{w}) \right)$$

(ii) If Z^{reg} is non-empty, then it is a Lagrangian subvariety of $\mathcal{M}(\mathbf{v}^1, \mathbf{w}), \mathcal{M}(\mathbf{v}^2, \mathbf{w})$ and each irreducible component of it has dimension exactly $\frac{1}{2} (\dim \mathcal{M}(\mathbf{v}^1, \mathbf{w}) + \dim \mathcal{M}(\mathbf{v}^2, \mathbf{w}))$

One special case of Steinberg variety is the Hecke correspondence constructed in [Nak98], section 5.

Let $\mathbf{e}^k$ be the dimension vector such that the k -th entry is 1 and the other be 0. Choose dimension vectors $\mathbf{v}^1, \mathbf{v}^2, \mathbf{w}$ such that $\mathbf{v}^2 = \mathbf{v}^1 + \mathbf{e}^k$. And let W, V^1, V^2 be vector spaces to the corresponding dimension vectors. The map $\mu^{-1}(0)^s \rightarrow \mathcal{M}$ is a principal $G_{\mathbf{v}}$ -bundle, then V the representation space of $G_{\mathbf{v}}$ can be regarded as holomorphic vector bundle. Thus we consider V^1, V^2 as vector bundles over $\mathcal{M}(\mathbf{v}, \mathbf{w}) \times \mathcal{M}(\mathbf{v}^2, \mathbf{w})$.

Take point $([\mathbf{p}^1, \mathbf{i}^1, \mathbf{j}^1], [\mathbf{p}^2, \mathbf{i}^2, \mathbf{j}^2])$, consider the sequence of vector bundles:

$$\bigoplus_{k \in Q_0} \text{Hom}(V_k^1, V_k^2) \xrightarrow{\sigma} \left(\left(t \bigoplus_{e \in Q_1} \text{Hom}(V_{s(e)}^1, V_{t(e)}^2) \oplus \bigoplus_{e \in Q_1} \text{Hom}(V_{s(e)}^1, V_{t(e)}^2) \right) \oplus t \text{Hom}(W, V^2) \oplus \text{Hom}(V^1, W) \right) \xrightarrow{\tau} t \text{Hom}(V^1, V^2) \oplus t\mathbf{k}$$

$$\sigma : \xi \mapsto (\mathbf{p}^2 \xi - \xi \mathbf{p}^1) \oplus (-\xi \mathbf{i}^1) \oplus \mathbf{j}^2 \xi; \quad \tau : C \oplus a \oplus b \mapsto (\varepsilon \mathbf{p}^2 C + \varepsilon C \mathbf{p}^1 + \mathbf{i}^2 b + a \mathbf{j}^1) \oplus (\text{Tr}(\mathbf{i}^1 b) + \text{Tr}(a \mathbf{j}^2))$$

where the notation $t^m V$ is for V tensor one-dimensional $\mathbb{G}_m$ -module of weight m . It is shown in [Nak98] that

Proposition 3.3.5. (i) The composition $\tau\sigma = 0$, so that the sequence given above is a complex.

(ii) The map σ is injective and τ is surjective, so that the sequence is exact.

(iii) The quotient $\text{Ker}(\tau)/\text{Im}(\sigma)$ is a vector bundle whose rank is $\frac{1}{2} (\dim \mathcal{M}(\mathbf{v}^1, \mathbf{w}) + \dim \mathcal{M}(\mathbf{v}^2, \mathbf{w}))$

Take s a section of $\text{Ker}(\tau)/\text{Im}(\sigma)$ given by $s = 0 \oplus (-\mathbf{i}^2) \oplus \mathbf{j}^1 \pmod{\text{Im} \sigma}$. It can be verified that $\tau(s) = 0$.

The point $([\mathbf{p}^1, \mathbf{i}^1, \mathbf{j}^1], [\mathbf{p}^2, \mathbf{i}^2, \mathbf{j}^2])$ is in the zero locus $Z(s)$ of s if and only if there exists $\xi \in \text{Hom}(V^1, V^2)$ such that

$$\xi \mathbf{p}^1 = \mathbf{p}^2 \xi, \quad \xi \mathbf{i}^1 = \mathbf{i}^2, \quad \mathbf{j}^1 = \mathbf{j}^2 \xi$$

This is equivalent to give a morphism between framed representations.

Definition 3.3.6. Call the zero locus $Z(s)$ the **Hecke correspondence**, denote it by $\mathfrak{B}_k(\mathbf{v}^2, \mathbf{w})$

We then consider the convolution algebra structure brought by (Borel-Moore) homology, see the Appendix A.

For three dimension vectors $\mathbf{v}^1, \mathbf{v}^2, \mathbf{v}^3$, we have $Z(\mathbf{v}^1, \mathbf{v}^2, \mathbf{w}) \circ Z(\mathbf{v}^2, \mathbf{v}^3, \mathbf{w}) \subseteq Z(\mathbf{v}^1, \mathbf{v}^3, \mathbf{w})$, and the convolution product then follows:

$$* : H_*^{\text{BM}} \left(Z(\mathbf{v}^1, \mathbf{v}^2, \mathbf{w}) \right) \otimes H_*^{\text{BM}} \left(Z(\mathbf{v}^2, \mathbf{v}^3, \mathbf{w}) \right) \rightarrow H_*^{\text{BM}} \left(Z(\mathbf{v}^1, \mathbf{v}^3, \mathbf{w}) \right)$$

Denote by $H_*^{\text{BM}}(\mathbf{w}) = \bigoplus_{\mathbf{v}^1, \mathbf{v}^2} H_*^{\text{BM}}(Z(\mathbf{v}^1, \mathbf{v}^2, \mathbf{w}))$. We omit the mark BM to simplify the notation.

Proposition 3.3.7. (i) *The convolution makes $H_*(\mathbf{w})$ admit an associative algebra structure.*

(ii) *The top degree part $H_{\text{top}}(\mathbf{w}) := \bigoplus_{\mathbf{v}^1, \mathbf{v}^2} H_{2 \dim_{\mathbb{C}}(Z(\mathbf{v}^1, \mathbf{v}^2, \mathbf{w}))} (Z(\mathbf{v}^1, \mathbf{v}^2, \mathbf{w}))$ is a subalgebra of $H_*(\mathbf{w})$ under convolution.*

Define elements:

$$E_k = \sum_{\mathbf{v}^2} \left[\mathfrak{B}_k(\mathbf{v}^2, \mathbf{w}) \right]; \quad F_k = \sum_{\mathbf{v}^2} (-1)^{\frac{1}{2}(\dim \mathcal{M}(\mathbf{v}^1, \mathbf{w}) - \dim \mathcal{M}(\mathbf{v}^2, \mathbf{w}))} \left[{}^t \left(\mathfrak{B}_k(\mathbf{v}^2, \mathbf{w}) \right) \right]$$

where ${}^t : \mathcal{M}(\mathbf{v}^1, \mathbf{w}) \times \mathcal{M}(\mathbf{v}^2, \mathbf{w}) \rightarrow {}^t : \mathcal{M}(\mathbf{v}^2, \mathbf{w}) \times \mathcal{M}(\mathbf{v}^1, \mathbf{w})$ the exchange of factors.

For further discussion, we introduce the modified enveloping algebra for $\mathfrak{g}(C)$, recall that $\mathcal{U}(\mathfrak{g}) = \mathcal{U}(\mathfrak{n}_-) \otimes \mathcal{U}(\mathfrak{h}) \otimes \mathcal{U}(\mathfrak{n}_+) = \mathcal{U}^- \otimes \mathcal{U}^0 \otimes \mathcal{U}^+$. The **modified enveloping algebra** is given by

$$\tilde{\mathcal{U}}(\mathfrak{g}) := \mathcal{U}^- \otimes \left(\bigoplus_{\lambda \in P} \mathbb{C} a_\lambda \right) \otimes \mathcal{U}^+$$

where the multiplication in $\tilde{\mathcal{U}}$ is determined by the rules

$$\begin{aligned} a_\lambda a_\mu &= \delta_{\lambda\mu} a_\lambda \\ e_k a_\lambda &= a_{\lambda+\alpha_k} e_k, \quad f_k a_\lambda = a_{\lambda-\alpha_k} f_k \\ (e_k f_l - f_l e_k) a_\lambda &= \delta_{kl} \langle \lambda, h_k \rangle a_\lambda \end{aligned}$$

Let $\Lambda_{\mathbf{w}} := \sum_{i \in Q_0} \mathbf{w}_i \Lambda_i$, where $\Lambda_i \in P$ are fundamental weights, and let $\alpha_{\mathbf{v}} := \sum_i \mathbf{v}_i \alpha_i$. Nakajima showed the following:

Theorem 3.3.8 ([Nak98], Theorem 9.4). *There exists the unique algebra homomorphism*

$$\Phi : \tilde{\mathcal{U}} \rightarrow \bigoplus_{\mathbf{v}^1, \mathbf{v}^2} H_{\text{top}} \left(Z(\mathbf{v}^1, \mathbf{v}^2; \mathbf{w}) \right)$$

such that

$$\begin{aligned} \Phi(a_\lambda) &= \begin{cases} [\Delta(\mathbf{v}, \mathbf{w})] & \text{if } \lambda = \Lambda_{\mathbf{w}} - \alpha_{\mathbf{v}} \\ 0 & \text{if there is no such } \mathbf{v} \end{cases} \\ \Phi(e_k a_\lambda) &= E_k \Phi(a_\lambda), \quad \Phi(a_\lambda f_k) = \Phi(a_\lambda) F_k \end{aligned}$$

Further, we could construct modules over $\tilde{\mathcal{U}}$. Choose $\mathbf{v}^0 \in \mathbb{Z}_+^{Q_0}$, and some point $x \in \mathcal{M}_0^{\text{reg}}(\mathbf{v}^0, \mathbf{w})$ if it is non-empty.

Theorem 3.3.9 ([Nak98] Theorem 10.2). *Assume that $\mathcal{M}_0^{\text{reg}}(\mathbf{v}^0, \mathbf{w})$ is non-empty, we have:*

(i) *The weight $\lambda := \Lambda_{\mathbf{w}} - \alpha_{\mathbf{v}^0}$ is dominant, hence $L(\lambda)$ is an integrable highest weight module.*

(ii) *For point $x \in \mathcal{M}_0^{\text{reg}}(\mathbf{v}^0, \mathbf{w})$, let $\mathcal{M}(\mathbf{v}, \mathbf{w})_x$ denote $\pi^{-1}(x) \subseteq \mathcal{M}(\mathbf{v}, \mathbf{w})$ for $\mathbf{v} \geq \mathbf{v}^0$, where π is the natural projection $\mathcal{M} \rightarrow \mathcal{M}_0$.*

Then $\bigoplus_{\mathbf{v} \geq \mathbf{v}^0} H_{\text{top}}(\mathcal{M}(\mathbf{v}, \mathbf{w})_x)$ has a $\tilde{\mathcal{U}}$ -module structure. Moreover, for each $H_{\text{top}}(\mathcal{M}(\mathbf{v}, \mathbf{w})_x)$, it is isomorphic to the weight space $L(\lambda)_{\Lambda_{\mathbf{w}} - \alpha_{\mathbf{v}}}$.

With the help of Theorem 3.3.9 and the Kac-Weyl formula 3.1, In [Hau10], T. Hausel successfully proved the conjecture of Kac that the Kac polynomial $A_Q(\alpha, q)$ has constant term equals to the multiplicity m_α of α .

Appendices

Appendix A

Borel-Moore Homology

In a general setting, we consider the topological space X that is locally compact (but not necessarily compact) with finitely many components and its homology and homotopy groups are also finitely generated (which implies X is of finite homotopy type of CW -complexes). Moreover, X can be embedded into a C^∞ manifold M and this embedding is closed. Let $\hat{X}$ denote one-point compactification, $\bar{X}$ arbitrary compactification.

Definition A.0.1. Define the **Borel-Moore homology** of X to be the relative homology group

$$H_*^{BM}(X) := H_*(\hat{X}, \text{pt}) \quad \text{of coefficient } \mathbb{C}$$

'pt' is the abstract point add to X to make compactification, H_* is the ordinary homology.

Remark. Several equivalent definitions of Borel-Moore homology are equivalent, we have

$$(i) H_*^{BM}(X) \simeq H_*(\bar{X}, \bar{X} \setminus X). \quad (ii) H_i^{BM}(X) \simeq H^{m-i}(M, M \setminus X).$$

If one wants to see the construction more algebraically from the chain complexes. The Borel-Moore chain complex $C_*^{BM}(X)$ has $\sum_{i=1}^{\infty} a_i \sigma_i$, $a_i \in \mathbb{C}$, σ_i simplex. We demand that for any compact subset $V \subseteq X$, there are only finite σ_i have nonzero intersection with D . This makes the formal sum locally finite. We take the boundary map ∂ the same as the one in chain of σ_i . And $H_*^{BM}(X) = H_*(C_*^{BM}(X), \partial)$.

Remark. From those constructions we find if X is compact, then its Borel-Moore homology coincides with ordinary homology. We next will see how things in ordinary homology fit Borel-Moore homology.

In our following discussion, to simplify the notations, we use H_* to denote the BM homology and H_*^{ord} for the ordinary one. And we mainly consider the case that X is a quasi-projective, complex variety.

Proposition A.0.2. Suppose X is a complex variety of dimension m (thus $\dim_{\mathbb{R}} X = 2m$). Set the fundamental class $[X] \in H_{2m}(X)$ to be $\sum_i [X_i]$, where X_i are irreducible components. Then $\{[X_i]\}$ forms a basis for $H_{2m}(X)$.

Consider $U \stackrel{\text{open}}{\subseteq} X$, there is a natural morphism $H_*(X) \rightarrow H_*(U)$ from the ordinary homology $H_*^{\text{ord}}(\bar{X}, \bar{X} \setminus X) \rightarrow H_*^{\text{ord}}(\bar{X}, \bar{X} \setminus U)$. If $F \stackrel{\text{closed}}{\subseteq} X$, let $U := X \setminus F$, then Poincaré duality $H_k(M) \simeq H^{n-k}(M)$ gives a long exact sequence

$$\dots \rightarrow H_d(F) \rightarrow H_d(X) \rightarrow H_d(U) \rightarrow H_{d-1}(F) \rightarrow \dots$$

Now assume X is a smooth, oriented manifold and $Z, \tilde{Z}$ closed submanifolds of X , then we have a bilinear pairing called **intersection pairing** $\frown: H_i(Z) \times H_j(\tilde{Z}) \rightarrow H_{i+j-n}(Z \cap \tilde{Z})$, $n = \dim_{\mathbb{R}} X$ which is obtained from

$$\frown: H^{n-i}(X, X \setminus Z) \times H^{n-j}(X, X \setminus \tilde{Z}) \rightarrow H^{2n-j-i}(X, (X \setminus Z) \cup (X \setminus \tilde{Z})) = H^{2n-j-i}(X, X \setminus (Z \cap \tilde{Z}))$$

the cup product of relative ordinary cohomology and Poincaré duality on each term.

Following the Künneth formula of relative ordinary homology

$$H_*^{\text{ord}}(\bar{X}_1, \bar{X}_1 \setminus X_1) \otimes H_*^{\text{ord}}(\bar{X}_2, \bar{X}_2 \setminus X_2) \simeq H_*^{\text{ord}}(\overline{X_1 \times X_2}, \overline{X_1 \times X_2} \setminus (\bar{X}_1 \times X_2 \cup X_1 \times \bar{X}_2))$$

we get the Künneth formula of BM homology, $\boxtimes : H_*(X_1) \otimes H_*(X_2) \rightarrow H_*(X_1 \times X_2)$.

Next we introduce two techniques called **specialization** and **convolution** which are important in the applications of Borel-Moore homology.

Consider a smooth manifold S of dimension d with a fixed base point o and let $S^* := S \setminus o$. For a map $\pi : Z \rightarrow S$, let $Z_{S'} := \pi^{-1}(S')$ for any subset $S' \subseteq S$ and Z_o denote $\pi^{-1}(o)$. Let (B, o) be an open neighbourhood of o that is diffeomorphic to $(\mathbb{R}^d, 0)$, o is mapped to 0 under this diffeomorphism. Let $\mathbb{R}_{>0} := (0, +\infty)$, $\mathbb{R}_{\geq 0} := [0, +\infty)$, $\mathbb{R}_{>0}^d := \mathbb{R}_{>0} \times \mathbb{R}^{d-1}$. Let $B_{>0} \subseteq B$ be the space corresponding to $\mathbb{R}_{>0}^d$ under the diffeomorphism and $I_{>0}$ (resp. $I_{\geq 0}$) $\subseteq B$ the space corresponding to $(0, +\infty) \times \{0\}$ (resp. $[0, +\infty) \times \{0\}$) $\subseteq \mathbb{R} \times \mathbb{R}^{d-1} = \mathbb{R}^d$.

Assume also $Z(S^*) \rightarrow S^*$ is a locally trivial fibration, and $Z(B_{>0}) \rightarrow B_{>0}$ is trivial fibration with fibre F , then

$$\begin{aligned} H_d(B_{>0}) &\simeq H_d(\mathbb{R}_{>0}^d) \simeq H_1(\mathbb{R}_{>0}) \simeq H_1(I_{>0}); \\ H_\bullet(Z(B_{>0})) &\simeq H_{\bullet-d}(F) \otimes H_d(B_{>0}) \simeq H_{\bullet-d}(F) \otimes H_1(I_{>0}) \simeq H_{\bullet-d+1}(Z(I_{>0})) \end{aligned}$$

We define the **specialization** to be the morphism $\lim_{S \rightarrow 0} : H_\bullet(Z(S^*)) \rightarrow H_{\bullet-d}(Z_o)$ from the following composition

$$H_\bullet(Z(S^*)) \rightarrow H_\bullet(Z(B_{>0})) \xrightarrow{\simeq} H_{\bullet-d+1}(Z(I_{>0})) \rightarrow H_{\bullet-d}(Z_o)$$

where the last morphism comes from the boundary map in the long exact sequence of the closed embedding $Z_o \hookrightarrow Z(I_{\geq 0})$.

Now let M_1, M_2, M_3 be three smooth, oriented manifolds, with $Z_{12} \stackrel{\text{closed}}{\subseteq} M_1 \times M_2$, and $Z_{23} \stackrel{\text{closed}}{\subseteq} M_2 \times M_3$. Define convolution $Z_{13} := Z_{12} \circ Z_{23} := \{(m_1, m_3) \in M_1 \times M_3 \mid \exists m_2 \in M_2, \text{ such that } (m_1, m_2) \in Z_{12}, (m_2, m_3) \in Z_{23}\}$. Let p_{ij} denote the projection $M_1 \times M_2 \times M_3 \rightarrow M_i \times M_j$. Moreover, assume $p_{13} : p_{12}^{-1}(Z_{12}) \cap p_{23}^{-1}(Z_{23}) \rightarrow Z_{12} \circ Z_{23}$ is proper. Define the **convolution** of homology $* : H_i(Z_{12}) \times H_j(Z_{23}) \rightarrow H_{i+j-2d}(Z_{12} \circ Z_{23})$, $(c, d) \mapsto c * d$. Here c, d are cycles in $H_i(Z_{12})$ and $H_j(Z_{23})$ respectively, and $c * d$ is given by

$$c * d = (p_{13})_*((c \boxtimes [M_3]) \frown (d \boxtimes [M_1])) = (p_{13})_*(p_{12}^*c \frown p_{23}^*d)$$

it is well-defined since $p_{13} : p_{12}^{-1}(Z_{12}) \cap p_{23}^{-1}(Z_{23}) \rightarrow Z_{12} \circ Z_{23}$ is proper by assumption.

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